LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - I

Matrices

Topics covered:

- Rank of a matrix: echelon form and normal form
- Cauchy-Binet formula (without proof)
- Inverse of a non-singular matrix by Gauss-Jordan method
- Systems of linear equations - homogeneous and non-homogeneous
- Gauss elimination method
- Jacobi and Gauss-Seidel iteration methods

Handwritten Notes - B.Tech (JNTUA R23)

1. Rank of a Matrix

Definition: the rank $r(A)$ of an $m \times n$ matrix A is the order of the largest non-zero minor (determinant) of A. Equivalently it is the number of NON-ZERO ROWS in its row echelon form.

- $r(A) = r(A^T)$, $r(A) \le \min(m, n)$, $r(A) = 0$ only for the zero matrix.
- For a square matrix of order n: $r(A) = n \Leftrightarrow |A| \neq 0$ (non-singular).
- Elementary row / column operations do NOT change the rank.
- $r(AB) \le \min(r(A), r(B))$.

Echelon form: a matrix is in (row) echelon form if (i) all zero rows are at the bottom, (ii) the first non-zero entry of each row (leading entry) lies to the right of the leading entry of the row above, so entries below a leading entry are zero.

Elementary operations: interchange $R_i \leftrightarrow R_j$; multiply a row by a non-zero constant $R_i \to kR_i$; add a multiple of one row to another $R_i \to R_i + kR_j$. (The same with columns C_i.)

Method: use the 1st row to make zeros below the first leading entry, then the 2nd row for the next column, and so on. Count the non-zero rows.

Problem 1: find the rank of $A = [1\ 2\ -1\ 3;\ 2\ 4\ 1\ -2;\ 3\ 6\ 3\ -7]$.

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 3 & 6 & 3 & -7 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Two non-zero rows remain $\Rightarrow r(A) = 2$.

Problem 2: find the rank of $A = [1\ 2\ 3;\ 2\ 4\ 7;\ 3\ 6\ 10]$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 3 & 6 & 10 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence $r(A) = 2$. (Check: $|A| = 0$ but the minor of order 2 formed by columns 1, 3 and rows 1, 2 is $1 \cdot 7 - 3 \cdot 2 = 1 \neq 0$.)

Problem 3: find the rank of the 4 x 4 matrix $A = [1\ 2\ 3\ 4;\ 2\ 3\ 4\ 5;\ 3\ 4\ 5\ 6;\ 4\ 5\ 6\ 7]$.

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -2 & -4 & -6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$$

$$\xrightarrow{R_4 \to R_4 - 4R_1} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -2 & -4 & -6 \\ 0 & -3 & -6 & -9 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & -6 & -9 \end{bmatrix} \xrightarrow{R_4 \to R_4 - 3R_2} \sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$r(A) = 2$ (the rows are in arithmetic progression, so every row is a combination of the first two).

Problem 4: find the rank of $A = [2\ 3\ -1\ -1;\ 1\ -1\ -2\ -4;\ 3\ 1\ 3\ -2;\ 6\ 3\ 0\ -7]$.

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 3R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 6 & 3 & 0 & -7 \end{bmatrix} \xrightarrow{R_4 \to R_4 - 6R_1} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} \xrightarrow{R_3 \to 5R_3 - 4R_2} \sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 0 & 33 & 22 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

$$\xrightarrow[\sim]{R4\to 5R4-9R2}\begin{bmatrix}1&-1&-2&-4\\0&5&3&7\\0&0&33&22\\0&0&33&22\end{bmatrix}\xrightarrow[\sim]{R4\to R4-R3}\begin{bmatrix}1&-1&-2&-4\\0&5&3&7\\0&0&33&22\\0&0&0&0\end{bmatrix}$$

Three non-zero rows (the last row has become zero) $\Rightarrow r(A) = 3$. The matrix is singular since $r < 4$.

Problem 5: find the rank of $A = [1\ 2\ 3;\ 2\ 4\ \lambda;\ 3\ 6\ 9]$ for different values of λ.

$$A=\begin{bmatrix}1&2&3\\2&4&\lambda\\3&6&9\end{bmatrix}\xrightarrow[\sim]{R2-2R1,\ R3-3R1}\begin{bmatrix}1&2&3\\0&0&\lambda-6\\0&0&0\end{bmatrix}$$

If $\lambda \neq 6$ the second row is non-zero $\Rightarrow r(A) = 2$. If $\lambda = 6$ the second row also vanishes $\Rightarrow r(A) = 1$.

2. Normal Form (Canonical Form)

Every $m \times n$ matrix of rank r can be reduced, by elementary ROW and COLUMN operations, to the normal form

$N = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$, and then $r(A) = r$. Equivalently $PAQ = N$ with P, Q non-singular.

Problem 6: reduce $A = [1\ 2\ 3\ 2;\ 2\ 3\ 5\ 1;\ 1\ 3\ 4\ 5]$ to normal form and find its rank.

$$A=\begin{bmatrix}1&2&3&2\\2&3&5&1\\1&3&4&5\end{bmatrix}\xrightarrow[\sim]{R2\to R2-2R1}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\1&3&4&5\end{bmatrix}\xrightarrow[\sim]{R3\to R3-R1}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\0&1&1&3\end{bmatrix}$$

$$\xrightarrow[\sim]{R3\to R3+R2}\begin{bmatrix}1&2&3&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow[\sim]{C2\to C2-2C1}\begin{bmatrix}1&0&3&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow[\sim]{C3\to C3-3C1}\begin{bmatrix}1&0&0&2\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}$$

$$\xrightarrow{C4\to C4-2C1}\sim\begin{bmatrix}1&0&0&0\\0&-1&-1&-3\\0&0&0&0\end{bmatrix}\xrightarrow{R2\to -R2}\sim\begin{bmatrix}1&0&0&0\\0&1&1&3\\0&0&0&0\end{bmatrix}\xrightarrow{C3\to C3-C2}\sim\begin{bmatrix}1&0&0&0\\0&1&0&3\\0&0&0&0\end{bmatrix}$$

$$\xrightarrow{C4\to C4-3C2}\sim\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&0\end{bmatrix}$$

This is $[I_2\ 0\ ;\ 0\ 0]\Rightarrow r(A)=2$.

Problem 7: find non-singular matrices P and Q such that PAQ is in normal form, for $A=[1\,1\,1\,;\,1\,2\,3\,;\,2\,3\,4]$.

Write $A=I\,A\,I$ and perform every row operation also on the LEFT identity and every column operation on the RIGHT identity:

$$R2-R1,\ R3-2R1,\ R3-R2\ ;\ C2-C1,\ C3-C1,\ C3-2C2$$

$$P=\begin{bmatrix}1&0&0\\-1&1&0\\-1&-1&1\end{bmatrix}\quad Q=\begin{bmatrix}1&-1&1\\0&1&-2\\0&0&1\end{bmatrix}\quad PAQ=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$$

So $PAQ=[I_2\ 0\ ;\ 0\ 0]$ and $r(A)=2$.

3. Cauchy–Binet Formula (without proof)

Let A be an $m\times n$ matrix and B an $n\times m$ matrix. Then $\det(AB)$ is the sum, over all choices S of m columns of A (equivalently the same m rows of B), of the product of the corresponding minors:

$$\boxed{\det(AB)=\Sigma\ \det(A_S)\cdot\det(B_S)}$$

(the sum is over all $C(n,m)$ selections S of m columns).

- If $m\le n$: $A_S=m\times m$ sub-matrix of A formed by the columns in S; $B_S=m\times m$ sub-matrix of B formed by the same rows.
- If $m>n$: $\det(AB)=0$ (because $r(AB)\le n<m$).
- If $m=n$ it reduces to $\det(AB)=\det A\cdot\det B$.

Problem 8: verify the formula for $A = [1\ 2\ 3; 4\ 5\ 6]$ and $B = [1\ 0; 0\ 1; 1\ 1]$.

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ 10 & 11 \end{bmatrix} \Rightarrow \det(AB) = -6$$

columns 1,2: $\det A_S = -3$, $\det B_S = 1 \rightarrow$ product $= -3$

columns 1,3: $\det A_S = -6$, $\det B_S = 1 \rightarrow$ product $= -6$

columns 2,3: $\det A_S = -3$, $\det B_S = -1 \rightarrow$ product $= 3$

Sum $= -6 = \det(AB)$ ✓ Cauchy–Binet verified.

Problem 9: for $A = [1\ 2; 3\ 4; 5\ 6]$ (3 × 2) and $B = [1\ 0\ 1; 2\ 1\ 0]$ (2 × 3) find $\det(AB)$.

Here $m = 3 > n = 2$, so by Cauchy–Binet $\det(AB) = 0$. (Direct check: AB is 3 × 3 of rank ≤ 2, hence singular.)

4. Inverse by Gauss–Jordan Method

A^{-1} exists only if $|A| \neq 0$. Write the augmented matrix $[A\,|\,I]$ and apply row operations ONLY until the left half becomes I; the right half is then A^{-1}. $[A\,|\,I] \sim [I\,|\,A^{-1}]$.

If a zero row appears in the left half, A is singular – no inverse.

Problem 10: find the inverse of $A = [1\ 2\ 3; 2\ 5\ 3; 1\ 0\ 8]$.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to -R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 3R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 + 3R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right] \xrightarrow{R_1 \to R_1 - 2R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Check: $A \cdot A^{-1} = I$ ✓.

Problem II: find the inverse of $A = [1\,1\,1;\ 1\,2\,3;\ 1\,4\,9]$.

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 3 & 0 & 1 & 0 \\ 1 & 4 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 1 & 4 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_1} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 3 & 8 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 3R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 2 & -3 & 1 \end{array}\right] \xrightarrow{R_3 \to (1/2)R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_3} \sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 0 & 3/2 & -1/2 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - R_2} \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -5/2 & 1/2 \\ 0 & 1 & 0 & -3 & 4 & -1 \\ 0 & 0 & 1 & 1 & -3/2 & 1/2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 3 & -5/2 & 1/2 \\ -3 & 4 & -1 \\ 1 & -3/2 & 1/2 \end{bmatrix}$$

Problem 12: is $A = [1\ 2\ ;\ 2\ 4]$ invertible?

$$[A|I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right]$$

A zero row appears in the left half ($|A| = 4 - 4 = 0$) $\Rightarrow$ A is singular, A^{-1} does not exist.

5. System of Linear Equations

$AX = B$ (m equations, n unknowns). Augmented matrix $[A \mid B] = [A\ B]$. Reduce it to echelon form (Gauss elimination). Rouché–Capelli test:

Condition	Nature of solution
$r(A) \neq r([A\|B])$	INCONSISTENT – no solution
$r(A) = r([A\|B]) = n$ (no. of unknowns)	CONSISTENT – UNIQUE solution
$r(A) = r([A\|B]) = r < n$	CONSISTENT – INFINITELY many solutions ($n - r$ free variables)

Homogeneous system $AX = 0$ is always consistent ($X = 0$ trivial solution). Non-trivial solutions exist $\Leftrightarrow$ $r(A) < n$ (for a square system $\Leftrightarrow |A| = 0$).

Gauss elimination: convert $[A \mid B]$ to upper-triangular (echelon) form by row operations, then solve by BACK-SUBSTITUTION.

Problem 13 (unique solution): $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 2 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + (-1/2)R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -3 & -9 \end{array}\right]$$

$r(A) = r([A|B]) = 3 = n \Rightarrow$ unique solution. Back-substitution: $-3z = -9 \to z = 3$; $-2y = -4 \to y = 2$; $x + 2 + 3 = 6 \to x = 1$.

$$\boxed{x = 1, y = 2, z = 3}$$

Problem 14 (infinitely many): $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 20$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 20 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 2 & 3 & 4 & 20 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & 2 & 8 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$r(A) = r([A|B]) = 2 < 3 = n \Rightarrow$ infinitely many solutions (one free variable). Put $z = K$: $y + 2z = 8 \to y = 8 - 2K$; $x = 6 - y - z = K - 2$.

$$\boxed{x = K - 2, y = 8 - 2K, z = K \ (K \text{ any real number})}$$

Problem 15 (inconsistent): $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 21$.

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 21 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 2 & 3 & 4 & 21 \end{array}\right] \xrightarrow{R_3 \to R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 1 & 2 & 9 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$r(A) = 2$ but $r([A|B]) = 3$ (last row reads $0 = 1$) $\Rightarrow$ the system is INCONSISTENT – no solution.

Problem 16: for what values of λ and μ does $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) infinitely many solutions?

$$[A|B] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right] \quad (R_2-R_1,\ R_3-R_1,\ R_3-R_2)$$

- $\lambda \neq 3$: $r(A) = r([A|B]) = 3 \rightarrow$ UNIQUE solution (for any μ).
- $\lambda = 3$ and $\mu \neq 10$: $r(A) = 2$, $r([A|B]) = 3 \rightarrow$ NO solution.
- $\lambda = 3$ and $\mu = 10$: $r(A) = r([A|B]) = 2 \rightarrow$ INFINITELY many solutions.

Problem 17 (homogeneous): solve $x + 3y - 2z = 0$, $2x - y + 4z = 0$, $x - 11y + 14z = 0$.

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 1 & -11 & 14 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

$r(A) = 2 < 3 \Rightarrow$ non-trivial solutions. Put $z = K$: $-7y + 8K = 0 \rightarrow y = 8K/7$; $x = -3y + 2z = -10K/7$. Taking $K = 7t$:

$$\boxed{(x, y, z) = t(-10, 8, 7)}$$

Problem 18: find λ for which $x + 2y + 3z = \lambda x$, $3x + y + 2z = \lambda y$, $2x + 3y + z = \lambda z$ has a non-trivial solution; solve it.

Non-trivial solution exists iff $|A - \lambda I| = 0$: expanding, $\lambda^3 - 3\lambda^2 - 9\lambda - 18 = 0$ i.e. $(\lambda - 6)(\lambda^2 + 3\lambda + 3) = 0$.

The only real value is $\lambda = 6$. For $\lambda = 6$: $-5x + 2y + 3z = 0$, $3x - 5y + 2z = 0$. By cross-multiplication $x : y : z = (4 + 15) : (9 + 10) : (25 - 6) = 19 : 19 : 19 \rightarrow x = y = z = K$.

$$\boxed{\lambda = 6,\ x = y = z = K}$$

6. Jacobi and Gauss-Seidel Iteration Methods

Used for large systems. The system must first be arranged so that the coefficient matrix is DIAGONALLY DOMINANT: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ (each diagonal element larger than the sum of the others in its row). Then both methods converge for any starting value.

For $a_1x + b_1y + c_1z = d_1$, etc. solve the i-th equation for the i-th unkno

$$\text{Jacobi}: x^{(K+1)} = \frac{d_1 - b_1y^{(K)} - c_1z^{(K)}}{a_1}, \; y^{(K+1)} \text{ uses } x^{(K)}, z^{(K)}, \ldots$$

$$\text{Gauss-Seidel}: y^{(K+1)} = \frac{d_2 - a_2x^{(K+1)} - c_2z^{(K)}}{b_2}, \ldots$$

(i.e. the NEWEST available values are used at once.)

Jacobi = simultaneous displacement (all old values); Gauss-Seidel = successive displacement (new values used immediately) – converges about twice as fast.

Problem 19: solve $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$ by Jacobi's method (start with $x = y = z = 0$).

Diagonally dominant ($10 > 2$, $10 > 3$, $10 > 4$) ✓. Iteration:

$$x = (12 - y - z)/10, \; y = (13 - 2x - z)/10, \; z = (14 - 2x - 2y)/10$$

K	x	y	z
0	0.0000	0.0000	0.0000
1	1.2000	1.3000	1.4000
2	0.9300	0.9200	0.9000
3	1.0180	1.0240	1.0300
4	0.9946	0.9934	0.9916
5	1.0015	1.0019	1.0024
6	0.9996	0.9995	0.9993

The values approach $x = y = z = 1$. (Exact solution 1, 1, 1.)

Problem 20: solve the same system by the Gauss-Seidel method.

K	x	y	z
0	0.0000	0.0000	0.0000
1	1.2000	1.0600	0.9480
2	0.9992	1.0054	0.9991
3	0.9996	1.0002	1.0001
4	1.0000	1.0000	1.0000

Already after 4 iterations Gauss-Seidel is almost at (1, 1, 1) – faster than Jacobi.

Problem 21: solve $20x + y - 2z = 17$, $3x + 20y - z = -18$, $2x - 3y + 20z = 25$ by Gauss-Seidel.

$x = (17 - y + 2z)/20$, $y = (-18 - 3x + z)/20$, $z = (25 - 2x + 3y)/20$

K	x	y	z
0	0.0000	0.0000	0.0000
1	0.8500	-1.0275	1.0109
2	1.0025	-0.9998	0.9998
3	1.0000	-1.0000	1.0000
4	1.0000	-1.0000	1.0000

Solution: $x \approx 1.000$, $y \approx -1.000$, $z \approx 1.000$.

Problem 22: solve $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$ by Gauss-Seidel (4 iterations).

K	x	y	z
0	0.0000	0.0000	0.0000
1	3.1481	3.5407	1.9132
2	2.4322	3.5720	1.9258
3	2.4257	3.5729	1.9260
4	2.4255	3.5730	1.9260

$x = 2.4255$, $y = 3.5730$, $z = 1.9260$ (to 4 decimals).

Problem 23: rearrange $x + 2y + 10z = 24$, $10x + y + z = 12$, $2x + 10y + z = 13$ so that the method converges.

Interchange so that the large coefficient is on the diagonal: $10x + y + z = 12$, $2x + 10y + z = 13$, $x + 2y + 10z = 24$. Now $10 > 2$, $10 > 3$, $10 > 3$ and Jacobi / Gauss-Seidel can be applied as above.

Important Questions - Unit I

1. Reduce a matrix to echelon form and find its rank (problems 1-5).
2. Reduce a matrix to normal form; find P and Q such that PAQ = normal form.
3. State the Cauchy-Binet formula and verify it for given matrices.
4. Find the inverse of a matrix by the Gauss-Jordan method.
5. Test consistency and solve a system by Gauss elimination (unique / infinite / none).
6. Find λ, μ for which a system has no / unique / infinitely many solutions.
7. Solve a homogeneous system; find non-trivial solutions.
8. Solve a system by Jacobi's and Gauss-Seidel iteration (3-4 iterations).

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UNIT - II

Eigenvalues, Eigenvectors & Orthogonal Transformation

Topics covered:

- Eigenvalues and eigenvectors and their properties
- Diagonalization of a matrix
- Cayley-Hamilton theorem (without proof)
- Inverse and powers of a matrix by Cayley-Hamilton theorem
- Quadratic forms and nature of quadratic forms
- Reduction of a quadratic form to canonical form by orthogonal transformation

Handwritten Notes - B.Tech (JNTUA R23)

1. Eigenvalues and Eigenvectors

For a square matrix A (order n), a non-zero vector X is an EIGENVECTOR and the scalar λ the corresponding EIGENVALUE if $AX = \lambda X$, i.e. $(A - \lambda I)X = 0$. Non-zero X exists iff $|A - \lambda I| = 0$:

$$|A - \lambda I| = 0 \text{ (characteristic equation)}$$

For a 3 x 3 matrix it expands to

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$$

S_1 = trace = $a_{11} + a_{22} + a_{33}$; S_2 = sum of the principal minors of order 2 (= sum of the co-factors of the diagonal elements); $|A|$ = determinant. For 2 x 2: $\lambda^2 - S_1\lambda + |A| = 0$.

Method: (1) form and solve the characteristic equation for λ; (2) for each λ solve $(A - \lambda I)X = 0$ by row reduction – the free variables give the eigenvector(s).

Properties of eigenvalues:

- Sum of eigenvalues = trace of A; product of eigenvalues = $|A|$.
- A and A^T have the same eigenvalues. A singular $\Leftrightarrow$ 0 is an eigenvalue.
- Triangular / diagonal matrix: eigenvalues are the diagonal elements.
- If λ is an eigenvalue of A then: $K\lambda$ for KA; λ^m for A^m; $1/\lambda$ for A^{-1}; $|A|/\lambda$ for adj A; $\lambda + K$ for $A + KI$; $p(\lambda)$ for $p(A)$.
- Eigenvectors belonging to DISTINCT eigenvalues are linearly independent.
- Real symmetric matrix: eigenvalues are real and eigenvectors of distinct eigenvalues are ORTHOGONAL.
- Eigenvalues of a skew-symmetric matrix are zero or purely imaginary; of an orthogonal matrix have modulus 1.

Problem 1: find the eigenvalues and eigenvectors of $A = [\,4\ 1\,;\,2\ 3\,]$.

$S_1 = 7$, $|A| = 10$: $\lambda^2 - 7\lambda + 10 = 0 \rightarrow (\lambda - 2)(\lambda - 5) = 0$. $\lambda = 2, 5$.

$$\lambda = 2: A - \lambda I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad (\lambda = 2)$$

$$\lambda = 5: A - \lambda I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (\lambda = 5)$$

Check: sum = 7 = trace ✓, product = 10 = $|A|$ ✓.

Problem 2: find the eigenvalues and eigenvectors of $A = [\,8\ -6\ 2\ ;\ -6\ 7\ -4\ ;\ 2\ -4\ 3\,]$.

$S_1 = 18$, $S_2 = (56 - 36) + (24 - 4) + (21 - 16) = 45$, $|A| = 0$.

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0 \rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$$

Eigenvalues $\lambda = 0, 3, 15$.

$$\lambda = 0: A - \lambda I = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad (\lambda = 0)$$

$$\lambda = 3 : A - \lambda I = \begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \quad (\lambda = 3)$$

$$\lambda = 15 : A - \lambda I = \begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$X = K \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad (\lambda = 15)$$

The eigenvectors are mutually orthogonal (A is symmetric) – e.g. $(1,2,2)\cdot(2,1,-2) = 2 + 2 - 4 = 0$ ✓.

Problem 3 (repeated eigenvalue): find the eigenvalues and eigenvectors of $A = [\,2\ 2\ 1\,;\,1\ 3\ 1\,;\,1\ 2\ 2\,]$.

$S_1 = 7$, $S_2 = 4 + 3 + 4 = 11$ … (cofactors of diagonal), $|A| = 5$.

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0 \rightarrow (\lambda - 1)^2(\lambda - 5) = 0$$

$\lambda = 1$ (repeated twice) and $\lambda = 5$.

$$\lambda = 1 : A - \lambda I = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$A - I$ has rank 1, so TWO free variables ($x_2 = s$, $x_3 = t$): $x_1 = -2s - t$.

$$X = s\begin{bmatrix}2\\-1\\0\end{bmatrix} + t\begin{bmatrix}1\\0\\-1\end{bmatrix} \quad (\lambda = 1)$$

$$\lambda = 5 : A - \lambda I = \begin{bmatrix}-3 & 2 & 1\\ 1 & -2 & 1\\ 1 & 2 & -3\end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix}1 & 0 & -1\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix}$$

$$X = k\begin{bmatrix}1\\1\\1\end{bmatrix} \quad (\lambda = 5)$$

For the repeated root we obtained two independent eigenvectors, so this A is diagonalizable.

Problem 4: eigenvalues of $A = [\,1\ 1\,;\,0\ 1\,]$. Triangular $\Rightarrow \lambda = 1, 1$.

$$\lambda = 1 : A - \lambda I = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} \overset{\text{row ops}}{\sim} \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$$

Only ONE independent eigenvector $(1, 0)^T$ for the double root $\Rightarrow$ this matrix is NOT diagonalizable (defective).

Problem 5 (using properties): the eigenvalues of A (3×3) are 1, 2, 3. Find the eigenvalues of (i) $A^2 + 2A + I$ (ii) A^{-1} (iii) adj A and $|A|$.

- (i) $p(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 \rightarrow 4, 9, 16$.
- (ii) $1/\lambda \rightarrow 1, 1/2, 1/3$.
- (iii) $|A| = 1 \cdot 2 \cdot 3 = 6$; $|A|/\lambda \rightarrow 6, 3, 2$ (so $|\text{adj } A| = 36$).

2. Diagonalization of a Matrix

A is DIAGONALIZABLE if there is a non-singular P with $P^{-1}AP = D$ (diagonal). P is the MODAL MATRIX whose columns are n linearly independent eigenvectors; D has the eigenvalues in the same order on its diagonal.

- A is diagonalizable ⇔ it has n linearly independent eigenvectors (always true if all eigenvalues are distinct, or if A is symmetric).
- Then $A = PDP^{-1}$ and $A^m = PD^mP^{-1}$ with $D^m = \text{diag}(\lambda_1^m, \lambda_2^m, \ldots)$.
- Real symmetric A can be diagonalised by an ORTHOGONAL matrix: $P^TAP = D$.

Problem 6: diagonalize $A = [\,1\ 1\ 3\,;\,1\ 5\ 1\,;\,3\ 1\ 1\,]$.

$$\lambda^3 - 7\lambda^2 + 36 = 0 \rightarrow (\lambda + 2)(\lambda - 3)(\lambda - 6) = 0$$

Eigenvectors: $\lambda = -2 \rightarrow (1, 0, -1)^T$; $\lambda = 3 \rightarrow (1, -1, 1)^T$; $\lambda = 6 \rightarrow (1, 2, 1)^T$ (found as in Problem 1–3).

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ -1 & 1 & 1 \end{bmatrix}, \quad P^{-1}AP = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

So $P^{-1}AP = \text{diag}(-2, 3, 6) = D$.

Problem 7: diagonalize $A = [\,4\ 1\,;\,2\ 3\,]$ and hence find A^5.

$$P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} 1/3 & -1/3 \\ 2/3 & 1/3 \end{bmatrix}$$

$$A^5 = PD^5P^{-1} = \begin{bmatrix} 2094 & 1031 \\ 2062 & 1063 \end{bmatrix}$$

Problem 8: diagonalize $A = [\,2\ 2\ 1\,;\,1\ 3\ 1\,;\,1\ 2\ 2\,]$ (repeated root).

$$P = \begin{bmatrix} -2 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Columns of P: two eigenvectors of $\lambda = 1$ and one of $\lambda = 5$ – three independent vectors ⇒ diagonalizable.

3. Cayley–Hamilton Theorem (without proof)

Every square matrix satisfies its own characteristic equation. If $\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$ is the characteristic equation of A then

$$\boxed{A^3 - S_1A^2 + S_2A - |A|I = 0}$$

Uses: (i) verify the theorem; (ii) find A^{-1}; (iii) find higher powers A^m and polynomials in A (reduce using the equation).

Problem 9 (verification): verify the theorem for $A = [1\ 2; 2\ 1]$.

$\lambda^2 - 2\lambda - 3 = 0$ ($S_1 = 2$, $|A| = -3$). To show $A^2 - 2A - 3I = 0$:

$$A^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, 2A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}, 3I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$A^2 - 2A - 3I = [5\ 4; 4\ 5] - [2\ 4; 4\ 2] - [3\ 0; 0\ 3] = [0\ 0; 0\ 0]$ ✓ Theorem verified.

Problem 10 (inverse): find A^{-1} using the theorem, for $A = [2\ 1\ 1; 0\ 1\ 0; 1\ 1\ 2]$.

$S_1 = 5$, $S_2 = (2 - 0) + (4 - 1) + (2 - 0) = 7$, $|A| = 3$.

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0 \Rightarrow A^3 - 5A^2 + 7A - 3I = 0$$

Multiply by A^{-1}: $A^2 - 5A + 7I - 3A^{-1} = 0 \Rightarrow$

$$\boxed{A^{-1} = \frac{1}{3}(A^2 - 5A + 7I)}$$

$$A_2 = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \quad A^{-1} = \tfrac{1}{3}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 3 & 0 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ 0 & 1 & 0 \\ -1/3 & -1/3 & 2/3 \end{bmatrix}$$

Problem 11 (polynomial in A): for the same A evaluate $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

Divide the polynomial by $\lambda^3 - 5\lambda^2 + 7\lambda - 3$:

$$= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + A^2 + A + I$$

The brackets are zero by Cayley–Hamilton, so the value is $A^2 + A + I$:

$$A^2 + A + I = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}$$

Problem 12 (powers): for $A = [1\ 2; 3\ 4]$ find A^4 using Cayley–Hamilton.

$\lambda^2 - 5\lambda - 2 = 0 \Rightarrow A^2 = 5A + 2I$. Then $A^3 = 5A^2 + 2A = 27A + 10I$; $A^4 = 27A^2 + 10A = 145A + 54I$.

$$A^4 = 145A + 54I = \begin{bmatrix} 199 & 290 \\ 435 & 634 \end{bmatrix}$$

(Check by direct multiplication $A^2 \cdot A^2$ ✓.)

4. Quadratic Forms and their Nature

A QUADRATIC FORM in n variables is a homogeneous polynomial of degree 2: $Q = X^T A X$ with A SYMMETRIC. Diagonal elements of A = coefficients of x_i^2; off-diagonal $a_{ij} = a_{ji} = \frac{1}{2}$ (coefficient of $x_i x_j$).

Problem 13: write the matrix of $Q = 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$.

$$Q = X^T A X, \quad A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

Canonical form: by a linear transformation $X = PY$, Q becomes a sum of squares $Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_r y_r^2$. Rank r = number of non-zero terms; index p = number of positive terms; signature = $2p - r$.

Nature	Eigenvalues of A	Principal minors
Positive definite	all $\lambda > 0$	all leading minors > 0
Negative definite	all $\lambda < 0$	minors alternate $-, +, -, \dots$
Positive semi-definite	all $\lambda \geq 0$, at least one $= 0$	≥ 0
Negative semi-definite	all $\lambda \leq 0$, at least one $= 0$	—
Indefinite	both positive and negative λ	otherwise

Problem 14: find the nature of $Q = x^2 + 2y^2 + 3z^2 + 2xy - 2yz$ by leading principal minors.

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

$$\Delta_1 = 1,\ \Delta_2 = 1,\ \Delta_3 = 2$$

All leading minors are positive ⇒ Q is POSITIVE DEFINITE.

Problem 15: nature of $Q = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ using eigenvalues.

Characteristic equation $\lambda^3 - 6\lambda^2 + 9\lambda = 0 \rightarrow \lambda(\lambda - 3)^2 = 0$: eigenvalues 0, 3, 3.

All $\lambda \geq 0$ and one is 0 ⇒ POSITIVE SEMI-DEFINITE ; canonical form $3y_2^2 + 3y_3^2$, rank 2, index 2, signature 2.

Problem 16: nature of $Q = x^2 + 4xy + y^2$.

$A = [1\ 2\ ;\ 2\ 1]$: $\lambda^2 - 2\lambda - 3 = 0 \rightarrow \lambda = 3, -1$. One positive, one negative ⇒ INDEFINITE. Canonical form $3y_1^2 - y_2^2$: rank 2, index 1, signature 0.

5. Reduction to Canonical Form by Orthogonal Transformation

Procedure for $Q = X^T AX$ (A real symmetric):

1. Find the eigenvalues $\lambda_1, \lambda_2, \lambda_3$ of A.
2. Find the eigenvectors; for DISTINCT λ they are already orthogonal – for a repeated λ choose orthogonal vectors (Gram–Schmidt).
3. Normalise each vector (divide by its length) and use them as the columns of the ORTHOGONAL matrix P ($P^TP = I$).
4. Put $X = PY$. Then $Q = Y^TDY = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$ (canonical form), with $P^TAP = D$.

Problem 17: reduce $Q = 6x^2 + 3y^2 - 4xy$ to canonical form by an orthogonal transformation.

$A = [\,6\ -2\ ;\ -2\ 3\,]$: $\lambda^2 - 9\lambda + 14 = 0 \rightarrow \lambda = 2, 7$.

$\lambda = 2$: $4x - 2y = 0 \rightarrow X_1 = (1, 2)^T$; $\lambda = 7$: $-x - 2y = 0 \rightarrow X_2 = (2, -1)^T$. ($X_1 \cdot X_2 = 0$ ✓). Normalising by $\sqrt{5}$:

$$P = \frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix},\ P^T A P = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\boxed{X = PY : Q = 2y_1^2 + 7y_2^2}$$

Nature: positive definite (both eigenvalues positive), rank 2, index 2, signature 2.

Problem 18: reduce $Q = 3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to canonical form by an orthogonal transformation.

Characteristic equation: $\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0 \rightarrow (\lambda - 2)(\lambda - 3)(\lambda - 6) = 0$.

$$\lambda = 2 : X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\quad \lambda = 3 : X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}\quad \lambda = 6 : X_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$X_1 \cdot X_2 = 0$, $X_2 \cdot X_3 = 0$, $X_1 \cdot X_3 = 0$ – mutually orthogonal. Lengths $\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$.

$$P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix},\ P^T A P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$\boxed{X = PY : Q = 2y_1^2 + 3y_2^2 + 6y_3^2}$$

Nature: POSITIVE DEFINITE; rank 3, index 3, signature 3.

Problem 19 (repeated eigenvalue): reduce $Q = 2x^2 + 2y^2 + 2z^2 - 2xy - 2yz - 2zx$ by an orthogonal transformation.

$\lambda = 0, 3, 3$. $\lambda = 0 : X_1 = (1, 1, 1)^T$. $\lambda = 3 : -x - y - z = 0$ (rank 1). Choose two ORTHOGONAL solutions: $X_2 = (1, -1, 0)^T$ and $X_3 = (1, 1, -2)^T$ ($X_2 \cdot X_3 = 0$ ✓, both $\perp X_1$).

$$P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}, P^T AP = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$X = PY : Q = 0 \cdot y_1^2 + 3y_2^2 + 3y_3^2$$

Positive semi-definite ; rank 2, index 2, signature 2.

Problem 20: reduce $Q = x^2 + 3y^2 + 3z^2 - 2yz$ to canonical form.

A = diag-block : eigenvalues 1, 2, 4. $X_1 = (1,0,0)^T$ ($\lambda = 1$) ; $\lambda = 2$: $y + \ldots \rightarrow$ $X_2 = (0, 1, 1)^T$; $\lambda = 4$: $X_3 = (0, 1, -1)^T$. Normalised:

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

$$X = PY : Q = y_1^2 + 2y_2^2 + 4y_3^2 \text{ (positive definite, rank 3, index 3)}$$

Important Questions – Unit II

1. Find eigenvalues and eigenvectors of a 2 x 2 / 3 x 3 matrix (distinct and repeated roots).
2. Properties of eigenvalues; find eigenvalues of A^2, A^{-1}, adj A, $A + kI$.
3. Diagonalize a matrix and find a power A^n.
4. State Cayley-Hamilton theorem; verify it; find A^{-1} and powers.
5. Evaluate a matrix polynomial using Cayley-Hamilton.
6. Find the matrix of a quadratic form; find its nature, rank, index and signature.
7. Reduce a quadratic form to canonical form by an orthogonal transformation.

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UNIT - III

Calculus - Mean Value Theorems

Topics covered:

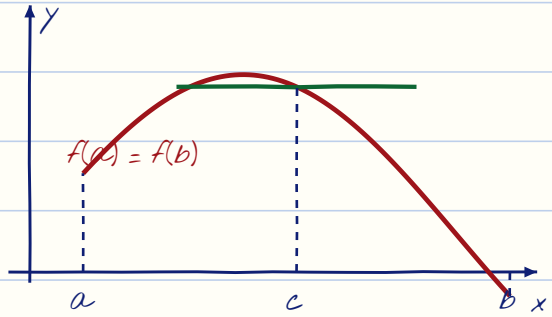
- Rolle's theorem and geometrical interpretation
- Lagrange's mean value theorem and geometrical interpretation
- Cauchy's mean value theorem
- Taylor's and Maclaurin's theorems with remainders (without proof)
- Problems and applications on the above theorems

Handwritten Notes - B.Tech (JNTUA R23)

1. Rolle's Theorem

Statement: if $f(x)$ is (i) continuous on the closed interval $[a, b]$, (ii) differentiable on the open interval (a, b) and (iii) $f(a) = f(b)$, then there exists at least one point $c \in (a, b)$ such that $f'(c) = 0$.

Geometrical meaning: if the ordinates at the end points are equal, the graph has at least one point where the tangent is parallel to the x-axis (horizontal).



Rolle: $f(a) = f(b)$

there is a point c where the tangent is PARALLEL to the x-axis : $f'(c) = 0$

Method to verify: (1) check the three conditions ; (2) solve $f'(c) = 0$; (3) show c lies in (a, b).

Problem 1: verify Rolle's theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

Polynomial – continuous and differentiable. $f(1) = 0 = f(3)$. $f'(x) = 2x - 4 = 0 \Rightarrow c = 2 \in (1, 3)$ ✓.

Problem 2: verify Rolle's theorem for $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$.

Continuous and differentiable everywhere. $f(\pi/4) = e^{\pi/4}(0) = 0$; $f(5\pi/4) = e^{5\pi/4}(-1/\sqrt{2} + 1/\sqrt{2}) = 0$.

$f'(x) = e^x(\sin x - \cos x) + e^x(\cos x + \sin x) = 2e^x \sin x = 0 \Rightarrow \sin x = 0$

$x = \pi \in (\pi/4, 5\pi/4) \Rightarrow c = \pi$. Rolle's theorem verified.

Problem 3: verify Rolle's theorem for $f(x) = e^{-x} \sin x$ on $[0, \pi]$.

$f(0) = 0 = f(\pi)$. $f'(x) = e^{-x}(\cos x - \sin x) = 0 \Rightarrow \tan x = 1 \Rightarrow c = \pi/4 \in (0, \pi)$ ✓.

Problem 4: verify Rolle's theorem for $f(x) = (x^2 + 3x)\, e^{-x/2}$ on $[-3, 0]$.

$f(-3) = 0 = f(0)$. $f'(x) = e^{-x/2}[\,2x + 3 - \tfrac{1}{2}(x^2 + 3x)\,] = \tfrac{1}{2}\,e^{-x/2}[\,4x + 6 - x^2 - 3x\,] = -\tfrac{1}{2}\,e^{-x/2}(x^2 - x - 6)$.

$f'(c) = 0 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x - 3)(x + 2) = 0 \Rightarrow c = -2 \in (-3, 0)$ ($c = 3$ is outside).

Problem 5: verify Rolle's theorem for $f(x) = \log\left[\,(x^2 + ab)\,/\,(x(a + b))\,\right]$ on $[a, b]$ $(0 < a < b)$.

$f(a) = \log\left[\,(a^2 + ab)/(a(a+b))\,\right] = \log 1 = 0$; $f(b) = \log\left[\,(b^2 + ab)/(b(a+b))\,\right] = \log 1 = 0$.

$$f'(x) = \frac{2x}{x^2 + ab} - \frac{1}{x} = 0 \Rightarrow 2x^2 = x^2 + ab \Rightarrow x^2 = ab$$

$c = \sqrt{(ab)}$, the geometric mean, lies in (a, b) ✓.

Problem 6 (when it fails): is Rolle's theorem applicable to $f(x) = 1 - x^{2/3}$ on $[-1, 1]$?

$f(-1) = 0 = f(1)$ and f is continuous, but $f'(x) = -(2/3)x^{-1/3}$ does not exist at $x = 0 \in (-1, 1)$. Condition (ii) fails ⇒ Rolle's theorem is NOT applicable (indeed $f'(x) \neq 0$ anywhere).

Problem 7: show that between two roots of $e^x \sin x = 1$ there is at least one root of $e^x \cos x = -1$.

Let $g(x) = e^{-x} - \sin x$. $g(x) = 0 \Leftrightarrow e^{-x} = \sin x \Leftrightarrow e^x \sin x = 1$. If a, b are two roots, $g(a) = g(b) = 0$ and g is differentiable, so by Rolle there is $c \in (a, b)$ with $g'(c) = 0$: $-e^{-c} - \cos c = 0 \Rightarrow e^c \cos c = -1$. Hence the claim.

Problem 8: verify Rolle's theorem for $f(x) = \cos x$ on $[-\pi/2, \pi/2]$.

$f(-\pi/2) = 0 = f(\pi/2)$; $f'(x) = -\sin x = 0 \Rightarrow x = 0 \in (-\pi/2, \pi/2) \Rightarrow c = 0$ ✓.

Problem 9: prove that $x^3 - 3x + K = 0$ cannot have two distinct roots in $[0, 1]$ (K any constant).

Suppose it had two roots $a < b$ in $[0, 1]$. $f(x) = x^3 - 3x + K$ satisfies Rolle's conditions, so $f'(c) = 3(c^2 - 1) = 0$ for some $c \in (a, b) \subset (0, 1)$. But $c^2 - 1 < 0$ for $0 < c < 1$ – contradiction. So at most one root lies in

$[0, 1]$.

2. Lagrange's Mean Value Theorem (LMVT)

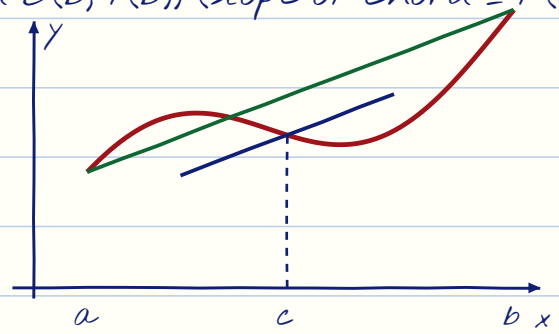
Statement: if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b), there exists $c \in (a, b)$ such that

$$f(b) - f(a) = (b - a)\, f'(c) \text{ i.e. } \frac{f(b) - f(a)}{b - a} = f'(c)$$

Alternative form: $f(a + h) = f(a) + h\, f'(a + \theta h)$, $0 < \theta < 1$ $(h = b - a)$.

Rolle's theorem is the special case $f(a) = f(b)$.

Geometrical meaning: there is a point on the curve where the tangent is PARALLEL to the chord joining the end points $A(a, f(a))$ and $B(b, f(b))$ (slope of chord $= f'(c)$).



Lagrange: chord AB

there is a point c where the tangent is PARALLEL to the chord AB

Problem 10: verify LMVT for $f(x) = x(x - 1)(x - 2)$ on $[0, 1/2]$.

$f(x) = x^3 - 3x^2 + 2x$; $f(0) = 0$; $f(1/2) = 1/2 \cdot (-1/2) \cdot (-3/2) = 3/8$. Slope of chord $= (3/8 - 0)/(1/2) = 3/4$.

$$f'(c) = 3c^2 - 6c + 2 = 3/4 \Rightarrow 12c^2 - 24c + 5 = 0$$

$$c = \frac{24 \pm \sqrt{(576 - 240)}}{24} = 1 \pm \frac{\sqrt{21}}{6}$$

$c = 1 - \sqrt{21}/6 \approx 0.2362 \in (0, 1/2)$ ✓ (the other root 1.76 is outside).

Problem 11: verify LMVT for $f(x) = \log x$ on $[1, e]$.

$f'(x) = 1/x$. Chord slope $= (\log e - \log 1)/(e - 1) = 1/(e - 1)$. $f'(c) = 1/c = 1/(e - 1)$

$\Rightarrow c = e - 1 \approx 1.718 \in (1, e)$ ✓.

Problem 12: find c of LMVT for $f(x) = x^3$ on $[1, 2]$.

$f(2) - f(1) = 8 - 1 = 7 = (2 - 1)\cdot 3c^2 \Rightarrow c^2 = 7/3 \Rightarrow c = \sqrt{(7/3)} \approx 1.5275 \in (1, 2)$ ✓.

Problem 13: find c for $f(x) = x^2 - 3x - 1$ on $[1117, 1317]$.

$f(b) - f(a) = (b^2 - a^2) - 3(b - a) = (b - a)(a + b - 3) \Rightarrow f'(c) = 2c - 3 = a + b - 3 \Rightarrow c = (a + b)/2 = 1217 \in (1117, 1317)$ ✓. (For ANY quadratic, c is the midpoint.)

Problem 14b: find c of LMVT for $f(x) = \sqrt{(x^2 - 4)}$ on $[2, 4]$.

$f(4) - f(2) = \sqrt{12} - 0 = 2\sqrt{3}$, chord slope $= \sqrt{3}$. $f'(x) = x/\sqrt{(x^2 - 4)} = \sqrt{3} \Rightarrow x^2 = 3x^2 - 12 \Rightarrow x^2 = 6 \Rightarrow c = \sqrt{6} \approx 2.449 \in (2, 4)$ ✓. (f is continuous at 2 and differentiable on the OPEN interval – enough for LMVT.)

Note: LMVT fails if the function is not differentiable inside, e.g. $f(x) = |x|$ on $[-1, 2]$: chord slope $= 1/3$ but $f' = \pm 1$ wherever it exists.

Applications – inequalities by LMVT:

Problem 15: prove $|\sin a - \sin b| \le |a - b|$.

Take $f(x) = \sin x$ on $[a, b]$: $\sin b - \sin a = (b - a)\cos c$. Since $|\cos c| \le 1$, $|\sin b - \sin a| \le |b - a|$.

Problem 16: prove $x/(1 + x) < \log(1 + x) < x$ for $x > 0$.

$f(t) = \log(1 + t)$ on $[0, x]$: $\log(1 + x) - 0 = x \cdot 1/(1 + c)$, $0 < c < x$. Since $1 < 1 + c < 1 + x$: $1/(1 + x) < 1/(1 + c) < 1$. Multiply by $x > 0$:

$$\frac{x}{1 + x} < \log(1 + x) < x$$

Problem 17: prove $(b - a)/(1 + b^2) < \tan^{-1}b - \tan^{-1}a < (b - a)/(1 + a^2)$, $0 < a < b$; hence $\pi/4 + 3/25 < \tan^{-1}(4/3) < \pi/4 + 1/6$.

$f(x) = \tan^{-1}x$: $\tan^{-1}b - \tan^{-1}a = (b - a)/(1 + c^2)$, $a < c < b$. As $a^2 < c^2 < b^2$, $1/(1 + b^2) < 1/(1 + c^2) < 1/(1 + a^2)$ – the result follows.

Take $a = 1$, $b = 4/3$: $b - a = 1/3$. Lower: $(1/3)/(1 + 16/9) = 3/25$; upper: $(1/3)/(1 + 1) = 1/6$. With $\tan^{-1}1 = \pi/4$:

$$\pi/4 + 3/25 < \tan^{-1}(4/3) < \pi/4 + 1/6$$

Problem 18: show that $e^x > 1 + x$ for $x > 0$ and that a function with $f'(x) = 0$ on an interval is constant.

(i) $f(t) = e^t$ on $[0, x]$: $e^x - 1 = x\,e^c > x$ since $e^c > 1$. (ii) For any $a < b$ in the interval, $f(b) - f(a) = (b - a)f'(c) = 0$, so $f(a) = f(b)$.

Problem 19: using LMVT prove $(b - a)/b < \log(b/a) < (b - a)/a$, $0 < a < b$.

$f(x) = \log x$ on $[a, b]$: $\log b - \log a = (b - a)/c$, $a < c < b$. Since $1/b < 1/c < 1/a$, multiply by $(b - a) > 0$: $(b - a)/b < \log(b/a) < (b - a)/a$. (For $a = 1$, $b = 1 + x$ this is the result of Problem 15.)

Problem 20: prove $\sin x < x$ for $x > 0$ and $\tan x > x$ for $0 < x < \pi/2$.

(i) $\sin x - \sin 0 = x \cos c \le x$ ($\cos c \le 1$; strict inequality since $\cos c < 1$ for $0 < c$). (ii) $\tan x - \tan 0 = x \sec^2 c > x$ since $\sec^2 c > 1$.

3. Cauchy's Mean Value Theorem

Statement: if $f(x)$ and $g(x)$ are (i) continuous on $[a, b]$, (ii) differentiable on (a, b), (iii) $g'(x) \neq 0$ for all x in (a, b), then there is $c \in (a, b)$ with

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Geometrically: for the curve $x = g(t)$, $y = f(t)$ there is a point where the tangent is parallel to the chord joining the end points. Taking $g(x) = x$ gives LMVT.

Problem 21: verify Cauchy's MVT for $f(x) = e^x$ and $g(x) = e^{-x}$ on $[a, b]$; show $c = (a + b)/2$.

$f'/g' = e^x/(-e^{-x}) = -e^{2c}$. LHS $= (e^b - e^a)/(e^{-b} - e^{-a}) = (e^b - e^a)\,e^{a+b}/(e^a - e^b) = -e^{a+b}$. So $e^{2c} = e^{a+b} \Rightarrow c = (a + b)/2 \in (a, b)$ ✓.

Problem 22: for $f(x) = \sqrt{x}$, $g(x) = 1/\sqrt{x}$ on $[a, b]$ show that c is the geometric mean $\sqrt{(ab)}$.

$f'/g' = [1/(2\sqrt{c})] / [-1/(2c^{3/2})] = -c$. LHS $= (\sqrt{b} - \sqrt{a})/(1/\sqrt{b} - 1/\sqrt{a}) = (\sqrt{b} - \sqrt{a})\sqrt{(ab)}/(\sqrt{a} - \sqrt{b}) = -\sqrt{(ab)}$. Hence $c = \sqrt{(ab)}$ ✓.

Problem 23: verify Cauchy's MVT for $f(x) = x^2$, $g(x) = x^3$ on $[1, 2]$.

LHS $= (4 - 1)/(8 - 1) = 3/7$. $f'/g' = 2c/(3c^2) = 2/(3c)$. $2/(3c) = 3/7 \Rightarrow c = 14/9 \approx 1.556 \in (1, 2)$ ✓.

Problem 24: verify Cauchy's MVT for $f(x) = \sin x$, $g(x) = \cos x$ on $[a, b]$.

LHS $= (\sin b - \sin a)/(\cos b - \cos a) = 2\cos((a+b)/2)\sin((b-a)/2) / [-2\sin((a+b)/2)\sin((b-a)/2)] = -\cot((a + b)/2)$. $f'/g' = \cos c/(-\sin c) = -\cot c. \Rightarrow c = (a + b)/2$ ✓.

Problem 25: using Cauchy's MVT show that $[f(b) - f(a)] / [g(b) - g(a)] = f'(c)/g'(c)$ reduces to LMVT for $g(x) = x$.

$g(b) - g(a) = b - a$ and $g'(c) = 1$, so $f(b) - f(a) = (b - a) f'(c)$.

Problem 26: for $f(x) = 1/x$, $g(x) = 1/x^2$ on $[a, b]$ find c.

$f'/g' = (-1/c^2)/(-2/c^3) = c/2$. LHS $= (1/b - 1/a)/(1/b^2 - 1/a^2) = ab/(a + b)$. So $c = 2ab/(a + b)$, the HARMONIC mean of a and b ✓.

Problem 27: for $f(x) = x$, $g(x) = x^2$ on $[a, b]$ find c.

LHS $= (b - a)/(b^2 - a^2) = 1/(a + b)$; $f'/g' = 1/(2c) \Rightarrow c = (a + b)/2$, the arithmetic mean ✓.

4. Taylor's and Maclaurin's Theorems (without proof)

Taylor's theorem: if f and its derivatives up to order $n-1$ are continuous on $[a, x]$ and $f^{(n)}$ exists on (a, x), then

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

$$R_n = \frac{(x-a)^n}{n!}f^{(n)}(c),\ a < c < x \quad \text{(Lagrange's form of remainder)}$$

Maclaurin's theorem: put $a = 0$:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{(n-1)}(0) + R_n$$

$$R_n = \frac{x^n}{n!}f^{(n)}(\theta x),\ 0 < \theta < 1$$

Standard Maclaurin series:

Function	Series	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + \dots$	all x
$\sin x$	$x - x^3/3! + x^5/5! - \dots$	all x
$\cos x$	$1 - x^2/2! + x^4/4! - \dots$	all x
$\log(1+x)$	$x - x^2/2 + x^3/3 - x^4/4 + \dots$	$-1 < x \le 1$
$(1+x)^m$	$1 + mx + m(m-1)x^2/2! + \dots$	$\lvert x\rvert < 1$
$\tan^{-1}x$	$x - x^3/3 + x^5/5 - \dots$	$\lvert x\rvert \le 1$
$\sinh x$, $\cosh x$	$x + x^3/3! + \dots$, $1 + x^2/2! + \dots$	all x

Problem 28: expand e^x by Maclaurin's theorem with Lagrange's remainder.

$f^{(K)}(x) = e^x \Rightarrow f^{(K)}(0) = 1$ for all K.

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!} + \frac{x^n}{n!} e^{\theta x}$$

Since $x^n/n! \to 0$ as $n \to \infty$, $R_n \to 0$ and the infinite series converges to e^x for every x.

Problem 29: expand $f(x) = x^3 - 2x^2 + 3x + 5$ in powers of $(x - 2)$.

$f(2) = 11$; $f' = 3x^2 - 4x + 3 \to f'(2) = 7$; $f'' = 6x - 4 \to f''(2) = 8$; $f''' = 6$.

$$f(x) = 11 + 7(x - 2) + \frac{8}{2!}(x - 2)^2 + \frac{6}{3!}(x - 2)^3$$

$$\boxed{= 11 + 7(x - 2) + 4(x - 2)^2 + (x - 2)^3}$$

Problem 30: expand $\log x$ in powers of $(x - 1)$ up to four terms.

$f = \log x$, $f' = 1/x$, $f'' = -1/x^2$, $f''' = 2/x^3$, $f'''' = -6/x^4$; at $x = 1$: $0, 1, -1, 2, -6$.

$$\boxed{\log x = (x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots}$$

Problem 31: expand $\sin x$ in powers of $(x - \pi/2)$.

Derivatives at $\pi/2$: $\sin = 1$, $\cos = 0$, $-\sin = -1$, $-\cos = 0$, $\sin = 1$ …

$$\sin x = 1 - \frac{(x - \pi/2)^2}{2!} + \frac{(x - \pi/2)^4}{4!} - \dots$$

Problem 32: find $\sin 31°$ (approximately) using Taylor's theorem about $\pi/6$.

$a = 30° = \pi/6$, $h = 1° = \pi/180 = 0.017453$. $f(a + h) = f(a) + h\,f'(a) + (h^2/2)\,f''(a) + (h^3/6)\,f'''(a)$ with $\sin \to \cos \to -\sin \to -\cos$:

$$\sin 31° \approx \tfrac{1}{2} + (\sqrt{3}/2)(0.017453) - \tfrac{1}{2} \cdot \frac{(0.017453)^2}{2} - (\sqrt{3}/2)\frac{(0.017453)^3}{6}$$

$$\boxed{= 0.5 + 0.015115 - 0.000076 - 0.0000008 \approx 0.51504}$$

Problem 33: find $\sqrt{4.1}$ approximately using Taylor's expansion of $\sqrt{x}$ about $x = 4$.

$f=\sqrt{x}$: $f(4)=2$, $f'=1/(2\sqrt{x})\to 1/4$, $f''=-1/(4x^{3/2})\to -1/32$, $f'''=3/(8x^{5/2})\to 3/256$.

$$\sqrt{4+h}=2+\frac{h}{4}-\frac{h^2}{64}+\frac{h^3}{512}-\dots$$

$h=0.1$: $2+0.025-0.000156+0.000002=2.02485$ (true value $2.024845\ldots$).

Problem 34: expand $e^x\sin x$ up to the term in x^5 (Maclaurin).

Multiply $e^x=1+x+x^2/2+x^3/6+x^4/24+x^5/120$ and $\sin x=x-x^3/6+x^5/120$:

$$\boxed{e^x\sin x=x+x^2+\frac{x^3}{3}+0\cdot x^4-\frac{x^5}{30}+\dots}$$

Problem 35: expand $\tan x$ up to x^5; and $e^{\sin x}$ up to x^4.

$$\tan x=x+\frac{x^3}{3}+\frac{2x^5}{15}+\dots$$

$$e^{\sin x}=1+x+\frac{x^2}{2}-\frac{x^4}{8}+\dots$$

Problem 36 (limits by series): evaluate $\lim (x-\sin x)/x^3$ and $\lim (\tan x-x)/x^3$ as $x\to 0$.

$x-\sin x=x^3/6-x^5/120+\dots\Rightarrow (x-\sin x)/x^3\to 1/6$. $\tan x-x=x^3/3+2x^5/15+\dots\Rightarrow(\tan x-x)/x^3\to 1/3$.

Problem 37 (applications with remainder): show that $\cos x\ge 1-x^2/2$ and $\sin x>x-x^3/6$ for $x>0$.

Maclaurin with 3 terms: $\cos x=1-x^2/2+(x^4/4!)\cos\theta x$ and $(x^4/24)\cos\theta x$ can be negative, but the 2-term form $\cos x=1-(x^2/2)\cos\theta x\ge 1-x^2/2$ for $x^2\ge 0$. Similarly $\sin x=x-x^3/6+(x^5/120)\cos\theta x$, and for $0<x<\pi/2$ the remainder is positive, so $\sin x>x-x^3/6$.

Problem 38: expand $\cos x$ by Maclaurin with remainder; and $(1+x)^{1/2}$ up to x^3.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{x^{2n}}{(2n)!}\cos(\theta x + n\pi)$$

$$\sqrt{(1+x)} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots$$

Problem 39: expand $\sin x$ in powers of $(x - \pi/4)$ up to the cubic term.

At $\pi/4$: $\sin = \cos = 1/\sqrt{2}$; $f' = \cos$, $f'' = -\sin$, $f''' = -\cos$.

$$\sin x = \frac{1}{\sqrt{2}}\left[1 + (x - \pi/4) - \frac{(x-\pi/4)^2}{2!} - \frac{(x-\pi/4)^3}{3!} + \dots\right]$$

Problem 40: expand e^x in powers of $(x - 1)$ and $\sec x$ up to x^4.

$$e^x = e\left[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \dots\right]$$

$$\sec x = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Problem 41 (more limits): evaluate $\lim (e^x - e^{-x} - 2x)/(x - \sin x)$ and $\lim (\cos x - 1 + x^2/2)/x^4$.

$e^x - e^{-x} - 2x = x^3/3 + \dots$, $x - \sin x = x^3/6 - \dots \Rightarrow$ ratio $\to (1/3)/(1/6) = 2$. $\cos x - 1 + x^2/2 = x^4/24 - \dots \Rightarrow$ limit $= 1/24$.

Problem 42 (error estimate): find the number of terms of the series of e^x needed to compute e to within 0.001.

$R_n = (1/n!)\,e^\theta < 3/n!$ for $x = 1$. Need $3/n! < 0.001 \Rightarrow n! > 3000 \Rightarrow n = 7$ ($7! = 5040$). So the first 7 terms (up to $1/6!$) give $e \approx 2.7181$ with error < 0.001.

Summary of the three mean value theorems:

Theorem	Conditions	Conclusion	Geometry
Rolle	f cont. on $[a,b]$, diff. on (a,b), $f(a) = f(b)$	$f'(c) = 0$	horizontal tangent
Lagrange	f cont. on $[a,b]$, diff. on (a,b)	$f(b) - f(a) = (b-a)\,f'(c)$	tangent ∥ chord
Cauchy	f, g cont. on $[a,b]$, diff. on (a,b), $g' \neq 0$	$[f(b)-f(a)]/[g(b)-g(a)] = f'(c)/g'(c)$	parametric curve
Taylor	f has derivatives up to order n	$f(x) = \Sigma\,(x-a)^k\,f^{(k)}(a)/k! + R_n$	polynomial approximation

Important Questions – Unit III

1. State and verify Rolle's theorem for given functions (problems 1–5); when does it fail?
2. State Lagrange's MVT, give its geometrical meaning, verify it and find c.
3. Prove inequalities using LMVT (problems 14–19).
4. State Cauchy's MVT and verify for pairs of functions.
5. State Taylor's and Maclaurin's theorems; expand e^x, $\sin x$, $\cos x$, $\log(1 + x)$.
6. Expand a given function in powers of $(x - a)$; approximate values ($\sin 31°$, $\sqrt{4.1}$).
7. Use Maclaurin series to evaluate limits and to expand products such as $e^x \sin x$, $\tan x$.

LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - IV

Partial Differentiation & Applications

Topics covered:

- Functions of several variables: continuity and differentiability
- Partial derivatives, total derivative, chain rule
- Directional derivative
- Taylor's and Maclaurin's series of functions of two variables
- Jacobians and functional dependence
- Maxima and minima of functions of two variables
- Method of Lagrange multipliers

Handwritten Notes - B.Tech (JNTUA R23)

1. Functions of Several Variables – Limit, Continuity

$z = f(x, y)$ is a function of two independent variables. LIMIT: $\lim f(x, y) = L$ as $(x, y) \to (a, b)$ means f approaches L along EVERY path to (a, b).

- If the limit has different values along two paths, the limit does NOT exist.
- f is CONTINUOUS at (a, b) if $\lim f(x, y) = f(a, b)$.
- Standard paths: $y = mx$, $y = mx^2$, $x = my^2$, $x = 0$, $y = 0$.

Problem 1: discuss the continuity of $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$ at the origin.

Along $y = mx$: $f = m\,x^2/(x^2(1 + m^2)) = m/(1 + m^2)$ – depends on m ($m = 0 \to 0$; $m = 1 \to 1/2$). The limit is not unique $\Rightarrow$ lim does not exist $\Rightarrow$ f is DISCONTINUOUS at $(0, 0)$.

Problem 2: examine the continuity of $f(x, y) = x^2y/(x^2 + y^2)$, $f(0, 0) = 0$ at $(0, 0)$.

$|f| = x^2|y|/(x^2 + y^2) \le |y|$ (since $x^2 \le x^2 + y^2$) $\to 0$ as $(x, y) \to (0, 0)$. So $\lim f = 0 = f(0, 0) \Rightarrow$ CONTINUOUS at the origin.

Problem 3: show that $\lim xy^2/(x^2 + y^4)$ does not exist at $(0, 0)$.

Along $y = mx$: $m^2x^3/(x^2 + m^4x^4) = m^2x/(1 + m^4x^2) \to 0$. Along $x = y^2$: $y^4/(y^4 + y^4) = 1/2$. Different limits $\Rightarrow$ the limit does not exist (checking only straight lines is not enough).

Differentiability: f is differentiable at (a, b) if $\Delta f = f_x\Delta x + f_y\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y$ with $\varepsilon_1, \varepsilon_2 \to 0$. If f_x and f_y exist and are CONTINUOUS at (a, b), f is differentiable there; differentiable $\Rightarrow$ continuous (converse false).

Problem 4: $f(x, y) = \sqrt{|xy|}$ at $(0, 0)$ – partial derivatives exist but f is not differentiable.

$f_x(0,0) = \lim [f(h,0) - f(0,0)]/h = 0$ and $f_y(0,0) = 0$. If f were differentiable, $\Delta f = 0\cdot h + 0\cdot K +$ small terms, i.e. $\sqrt{|hK|} = o(\sqrt{(h^2+K^2)})$; but along $h = K$: $\sqrt{(h^2)} = |h| = (1/\sqrt{2})\sqrt{(2h^2)}$ – a constant multiple of the distance, not smaller. Hence NOT differentiable.

2. Partial Derivatives

$f_x = \partial f/\partial x$ is the derivative with respect to x treating y as constant (and vice-versa). Second order: $f_{xx}, f_{yy}, f_{xy} = \partial^2 f/\partial y \partial x, f_{yx}$. If f_{xy}, f_{yx} are continuous then $f_{xy} = f_{yx}$ (Schwarz / Clairaut).

Problem 5: for $u = x^4 + y^4 - 4x^2y^2$ find $u_x, u_y, u_{xx}, u_{yy}, u_{xy}$.

$$u_x = 4x^3 - 8xy^2 \,,\; u_y = 4y^3 - 8x^2y$$

$$u_{xx} = 12x^2 - 8y^2 \,,\; u_{yy} = 12y^2 - 8x^2 \,,\; u_{xy} = u_{yx} = -16xy$$

Problem 6: if $u = \log(x^2 + y^2)$ show that $u_{xx} + u_{yy} = 0$ (Laplace equation).

$$u_x = \frac{2x}{x^2+y^2} \,,\; u_{xx} = \frac{2(y^2 - x^2)}{(x^2+y^2)^2} \,,\; u_{yy} = \frac{2(x^2 - y^2)}{(x^2+y^2)^2}$$

Adding: $u_{xx} + u_{yy} = 0$ ✓.

Problem 7: if $u = \tan^{-1}(y/x)$ show that $u_{xx} + u_{yy} = 0$.

$$u_x = \frac{-y}{x^2+y^2} \,,\; u_y = \frac{x}{x^2+y^2}$$

$$u_{xx} = \frac{2xy}{(x^2+y^2)^2} \,,\; u_{yy} = \frac{-2xy}{(x^2+y^2)^2} \Rightarrow u_{xx} + u_{yy} = 0 \text{ ✓}$$

Problem 8: if $u = x^y$ show that $u_{xy} = u_{yx}$.

$u_x = y\,x^{y-1} \Rightarrow u_{xy} = x^{y-1} + y\,x^{y-1}\log x = x^{y-1}(1 + y\log x)$. $u_y = x^y \log x \Rightarrow u_{yx} = y\,x^{y-1}\log x + x^y\cdot(1/x) = x^{y-1}(1 + y\log x)$. Equal ✓.

Problem 9: if $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $(\partial/\partial x + \partial/\partial y + \partial/\partial z)\,u = 3/(x+y+z)$ and $(\partial/\partial x + \partial/\partial y + \partial/\partial z)^2\,u = -9/(x+y+z)^2$.

Let $S = x^3 + y^3 + z^3 - 3xyz$. Sum of the three partials of $S = 3(x^2 - yz) + 3(y^2 - zx) + 3(z^2 - xy) = 3(x^2 + y^2 + z^2 - xy - yz - zx) = 3S/(x+y+z)$ [since

$S = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)]$.

$$(u_x+u_y+u_z) = \frac{3S/(x+y+z)}{S} = \frac{3}{x+y+z}$$

Applying the operator once more (each derivative of $1/(x+y+z)$ is $-1/(x+y+z)^2$):

$$\boxed{(\partial_x+\partial_y+\partial_z)\frac{3}{x+y+z} = -\frac{9}{(x+y+z)^2}} \checkmark$$

Euler's theorem on homogeneous functions: if $f(x,y)$ is homogeneous of degree n, then $x f_x + y f_y = n f$. Further $x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1) f$.

Problem 10: if $u = \sin^{-1}[(x^2+y^2)/(x+y)]$ show that $x u_x + y u_y = \tan u$.

$\sin u = (x^2+y^2)/(x+y) =: f$, homogeneous of degree 1. By Euler: $x f_x + y f_y = f$. But $f = \sin u \Rightarrow f_x = \cos u \cdot u_x$, $f_y = \cos u \cdot u_y$:

$$\boxed{\cos u\,(x u_x + y u_y) = \sin u \Rightarrow x u_x + y u_y = \tan u}$$

Problem 11: verify Euler's theorem for $u = x^3+y^3+3x^2y$ (degree 3).

$u_x = 3x^2+6xy$, $u_y = 3y^2+3x^2$. $x u_x + y u_y = 3x^3+6x^2y+3y^3+3x^2y = 3(x^3+y^3+3x^2y) = 3u$ ✓.

3. Total Derivative and Chain Rule

Total differential: $dz = (\partial z/\partial x)\,dx + (\partial z/\partial y)\,dy$.

Total derivative: if $z = f(x,y)$ with $x = x(t)$, $y = y(t)$:

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}}$$

Chain rule (two parameters): $z = f(x,y)$, $x = x(u,v)$, $y = y(u,v)$:

$\partial z/\partial u = z_x x_u + z_y y_u$, $\partial z/\partial v = z_x x_v + z_y y_v$.

Implicit function: if $f(x,y) = c$ defines $y(x)$: $dy/dx = -f_x/f_y$.

Problem 12: if $z = x^2y + xy^2$ with $x = at^2$, $y = 2at$, find dz/dt.

$z_x = 2xy + y^2$, $z_y = x^2 + 2xy$; $dx/dt = 2at$, $dy/dt = 2a$.

$$\frac{dz}{dt} = (2xy + y^2)(2at) + (x^2 + 2xy)(2a)$$

Substituting $x = at^2$, $y = 2at$: $2xy + y^2 = 4a^2t^3 + 4a^2t^2$; $x^2 + 2xy = a^2t^4 + 4a^2t^3$.

$$\frac{dz}{dt} = 8a^3t^4 + 8a^3t^3 + 2a^3t^4 + 8a^3t^3 = 10a^3t^4 + 16a^3t^3$$

Problem 13: if $u = x^2 + y^2 + z^2$ and $x = e^t$, $y = e^t \sin t$, $z = e^t \cos t$, find du/dt.

Directly: $u = e^{2t}(1 + \sin^2 t + \cos^2 t) = 2e^{2t} \Rightarrow du/dt = 4e^{2t}$. By the chain rule: $2x \cdot e^t + 2y \cdot e^t(\sin t + \cos t) + 2z \cdot e^t(\cos t - \sin t) = 4e^{2t}$ ✓.

Problem 14: if $z = f(x, y)$, $x = u \cos \alpha - v \sin \alpha$, $y = u \sin \alpha + v \cos \alpha$ show that $z_x^2 + z_y^2 = z_u^2 + z_v^2$.

$x_u = \cos \alpha$, $x_v = -\sin \alpha$, $y_u = \sin \alpha$, $y_v = \cos \alpha$. $z_u = z_x \cos \alpha + z_y \sin \alpha$, $z_v = -z_x \sin \alpha + z_y \cos \alpha$. Squaring and adding:

$$z_u^2 + z_v^2 = z_x^2(\cos^2\alpha + \sin^2\alpha) + z_y^2(\sin^2\alpha + \cos^2\alpha) = z_x^2 + z_y^2 \checkmark$$

Problem 15: if $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, show that $z_x^2 + z_y^2 = z_r^2 + (1/r^2) z_\theta^2$.

$z_r = z_x \cos \theta + z_y \sin \theta$; $z_\theta = -r z_x \sin \theta + r z_y \cos \theta$. Then $z_r^2 + z_\theta^2/r^2 = (z_x \cos \theta + z_y \sin \theta)^2 + (-z_x \sin \theta + z_y \cos \theta)^2 = z_x^2 + z_y^2$ ✓.

Problem 16: find dy/dx if $x^3 + y^3 = 3axy$.

$f_x = 3x^2 - 3ay$, $f_y = 3y^2 - 3ax$.

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{ay - x^2}{y^2 - ax}$$

4. Directional Derivative

The derivative of f in the direction of a unit vector $\hat{u} = (l, m, n)$:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = l f_x + m f_y + n f_z \text{ where } \nabla f = (f_x, f_y, f_z)$$

- Greatest rate of increase of f at a point is $|\nabla f|$, in the direction of ∇f.
- Smallest (most negative) rate is $-|\nabla f|$ in the direction $-\nabla f$. Along a level curve $D_{\hat{u}} f = 0$.
- The vector $\hat{u}$ must be a UNIT vector – divide a given direction by its length first.

Problem 17: find the directional derivative of $f = x^2 + xy + y^2$ at $(1, 2)$ in the direction of the vector $i + j$.

$\nabla f = (2x + y, x + 2y) = (4, 5)$ at $(1, 2)$. $\hat{u} = (1, 1)/\sqrt{2}$.

$$D_{\hat{u}} f = (4 + 5)/\sqrt{2} = 9/\sqrt{2} \approx 6.364$$

Maximum rate of change $= |\nabla f| = \sqrt{(16 + 25)} = \sqrt{41}$ in the direction $(4, 5)$.

Problem 18: find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2i - j - 2K$.

$\nabla f = (2xyz + 4z^2, x^2z, x^2y + 8xz) = (8, -1, -10)$ at $(1, -2, -1)$. $|2i - j - 2K| = 3 \Rightarrow \hat{u} = (2, -1, -2)/3$.

$$D_{\hat{u}} f = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

Problem 19: in what direction from $(1, 1)$ is the directional derivative of $f = x^2 - y^2 + 2xy$ a maximum? find its value.

$\nabla f = (2x + 2y, -2y + 2x) = (4, 0)$ at $(1, 1)$. Direction i (along the positive x-axis), maximum value $|\nabla f| = 4$.

5. Taylor's and Maclaurin's Series – Two Variables

$$f(x,y) = f(a,b) + [(x-a)f_x + (y-b)f_y]$$

$$+ \frac{1}{2!}[(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}] + \dots$$

(all derivatives evaluated at (a, b)). With $h = x - a$, $K = y - b$ the n-th term is $\frac{1}{n!}(h\,\partial/\partial x + K\,\partial/\partial y)^n f$. Maclaurin's series: $(a, b) = (0, 0)$.

Problem 20: expand $e^x \sin y$ in powers of x and y up to the third degree.

Derivatives at $(0, 0)$: $f = 0$; $f_x = 0$; $f_y = 1$; $f_{xx} = 0$; $f_{xy} = 1$; $f_{yy} = 0$; $f_{xxy} = 1$; $f_{yyy} = -1$; others 0.

$$e^x \sin y = y + xy + \frac{x^2 y}{2} - \frac{y^3}{6} + \dots$$

(Quick check: multiply the one-variable series $(1 + x + x^2/2 + \dots)(y - y^3/6 + \dots)$.)

Problem 21: expand $e^x \log(1 + y)$ up to the third degree.

$$e^x \log(1+y) = \left(1 + x + \frac{x^2}{2}\right)\left(y - \frac{y^2}{2} + \frac{y^3}{3}\right) = y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3} + \dots$$

Problem 22: expand $f(x, y) = x^2 y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ up to the third degree.

$f(1, -2) = -2 - 6 - 2 = -10$; $f_x = 2xy \to -4$; $f_y = x^2 + 3 \to 4$; $f_{xx} = 2y \to -4$; $f_{xy} = 2x \to 2$; $f_{yy} = 0$; $f_{xxy} = 2$; others 0.

$$f = -10 + [-4h + 4K] + \frac{1}{2}[-4h^2 + 4hK] + \frac{1}{6}[3 \cdot 2\, h^2 K]$$

$$= -10 - 4h + 4K - 2h^2 + 2hK + h^2 K \quad (h = x - 1,\ K = y + 2)$$

Problem 23: expand $\tan^{-1}(y/x)$ about $(1, 1)$ up to the second degree.

$f(1,1) = \pi/4$; $f_x = -y/(x^2+y^2) = -1/2$; $f_y = x/(x^2+y^2) = 1/2$; $f_{xx} = 2xy/(x^2+y^2)^2 = 1/2$; $f_{xy} = (y^2 - x^2)/(x^2+y^2)^2 = 0$; $f_{yy} = -1/2$.

$$\tan^{-1}(y/x) = \frac{\pi}{4} - \frac{x-1}{2} + \frac{y-1}{2} + \frac{(x-1)^2}{4} - \frac{(y-1)^2}{4} + \dots$$

Problem 24: Maclaurin series of $\cos x \cos y$ up to the fourth degree.

$$\cos x \cos y = 1 - \frac{x^2+y^2}{2} + \frac{x^2y^2}{4} + \frac{x^4+y^4}{24} + \dots$$

6. Jacobians and Functional Dependence

For $u = u(x,y)$, $v = v(x,y)$ the JACOBIAN is the determinant

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{bmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{bmatrix}$$

- $J \cdot J' = 1$ where $J' = \partial(x,y)/\partial(u,v)$ (inverse transformation).
- Chain rule: $\partial(u,v)/\partial(r,s) = \partial(u,v)/\partial(x,y) \cdot \partial(x,y)/\partial(r,s)$.
- Functional dependence: u, v (or u, v, w) are functionally DEPENDENT $\Leftrightarrow J = 0$; independent $\Leftrightarrow J \neq 0$.

Problem 25: find $\partial(x,y)/\partial(r,\theta)$ for $x = r\cos\theta$, $y = r\sin\theta$; and $\partial(r,\theta)/\partial(x,y)$.

$$J = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} = r\cos^2\theta + r\sin^2\theta = r \text{ , so } \partial(r,\theta)/\partial(x,y) = 1/r$$

Problem 26: $x = u(1-v)$, $y = uv$: find $\partial(x,y)/\partial(u,v)$.

$$J = \begin{bmatrix} 1-v & -u \\ v & u \end{bmatrix} = u(1-v) + uv = u$$

Problem 27: cylindrical and spherical coordinates: find the Jacobians.

Cylindrical ($x = r\cos\theta, y = r\sin\theta, z = z$): $\partial(x, y, z)/\partial(r, \theta, z) = r$

Spherical: $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$:

$$\boxed{\partial(x, y, z)/\partial(r, \theta, \varphi) = r^2\sin\theta}$$

Problem 28: if $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$, find $\partial(u, v)/\partial(r, \theta)$.

$\partial(u, v)/\partial(x, y) = \det\begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix} = 4x^2 + 4y^2 = 4r^2$. $\partial(x, y)/\partial(r, \theta) = r$.

$$\boxed{\partial(u, v)/\partial(r, \theta) = 4r^2 \cdot r = 4r^3}$$

Problem 29: show that $u = x + y + z$, $v = xy + yz + zx$, $w = x^3 + y^3 + z^3 - 3xyz$ are functionally dependent and find the relation.

Rows of the Jacobian: $(1, 1, 1)$; $(y + z, z + x, x + y)$; $(3(x^2 - yz), 3(y^2 - zx), 3(z^2 - xy))$. Since $w = u^3 - 3uv$ (shown below), Row 3 $= (3u^2 - 3v)\cdot$Row 1 $- 3u\cdot$Row 2, so the rows are linearly dependent and the determinant $J = 0 \Rightarrow$ DEPENDENT.

Relation: $w = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$ and $x^2 + y^2 + z^2 = u^2 - 2v$, so

$$\boxed{w = u(u^2 - 3v) = u^3 - 3uv}$$

Problem 30: prove $u = (x + y)/(1 - xy)$ and $v = \tan^{-1}x + \tan^{-1}y$ are dependent.

$u_x = (1 + y^2)/(1 - xy)^2$, $u_y = (1 + x^2)/(1 - xy)^2$; $v_x = 1/(1 + x^2)$, $v_y = 1/(1 + y^2)$. $J = u_xv_y - u_yv_x = [(1 + y^2)/(1 + y^2) - (1 + x^2)/(1 + x^2)]/(1 - xy)^2 = 0$. Relation: $\tan v = (x + y)/(1 - xy) = u$.

Problem 31: are $u = xy/z$, $v = yz/x$, $w = zx/y$ independent?

$J = 4$ (a non-zero constant) $\Rightarrow u, v, w$ are functionally independent. (Indeed $uv = y^2$, $vw = z^2$, $wu = x^2$ can be solved for x, y, z.)

7. Maxima and Minima of $f(x, y)$

Working rule:

1. Solve $f_x = 0$, $f_y = 0$ – the solutions are the STATIONARY (critical) points.
2. At each such point compute $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$.
3. $rt - s^2 > 0$ and $r < 0 \rightarrow$ MAXIMUM; $rt - s^2 > 0$ and $r > 0 \rightarrow$ MINIMUM.
4. $rt - s^2 < 0 \rightarrow$ SADDLE point (neither max nor min); $rt - s^2 = 0 \rightarrow$ doubtful, further study.

Problem 32: find the extreme values of $f = x^3 + y^3 - 3axy$ $(a > 0)$.

$f_x = 3x^2 - 3ay = 0$, $f_y = 3y^2 - 3ax = 0 \Rightarrow y = x^2/a$ and $x^4 = a^3x \Rightarrow x = 0$ or $x = a$. Points $(0, 0)$ and (a, a).

$r = 6x$, $s = -3a$, $t = 6y$. At $(0, 0)$: $rt - s^2 = -9a^2 < 0 \rightarrow$ SADDLE. At (a, a): $rt - s^2 = 36a^2 - 9a^2 = 27a^2 > 0$ and $r = 6a > 0 \rightarrow$ MINIMUM.

$$f_{min} = f(a, a) = a^3 + a^3 - 3a^3 = -a^3$$

Problem 33: find the minimum of $f = x^2 + y^2 + 6x + 12$.

$f_x = 2x + 6 = 0 \rightarrow x = -3$; $f_y = 2y = 0 \rightarrow y = 0$. $r = 2$, $s = 0$, $t = 2$: $rt - s^2 = 4 > 0$ and $r > 0 \rightarrow$ MINIMUM at $(-3, 0)$, value $f = 9 - 18 + 12 = 3$.

Problem 34: find the maximum of $f = x^3y^2(1 - x - y)$.

$f = x^3y^2 - x^4y^2 - x^3y^3$; $f_x = x^2y^2(3 - 4x - 3y) = 0$, $f_y = x^3y(2 - 2x - 3y) = 0$.

For $x, y \neq 0$: $4x + 3y = 3$ and $2x + 3y = 2 \Rightarrow x = 1/2$, $y = 1/3$.

Exact values: $r = -1/9$, $s = -1/12$, $t = -1/8$; $rt - s^2 = 1/144 > 0$ and $r < 0 \Rightarrow$ MAXIMUM.

$$f_{max} = (1/2)^3 (1/3)^2 (1 - 1/2 - 1/3) = \frac{1}{8} \cdot \frac{1}{9} \cdot \frac{1}{6} = \frac{1}{432}$$

Problem 35: show that $f = \sin x + \sin y + \sin(x + y)$ has a maximum at $(\pi/3, \pi/3)$.

$f_x = \cos x + \cos(x+y) = 0$, $f_y = \cos y + \cos(x+y) = 0 \Rightarrow \cos x = \cos y \Rightarrow x = y$; then $\cos x + \cos 2x = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0 \Rightarrow \cos x = 1/2 \Rightarrow x = y = \pi/3$.

$r = -\sin x - \sin(x+y) = -\sqrt{3}/2 - \sqrt{3}/2 = -\sqrt{3}$; $s = -\sin(x+y) = -\sqrt{3}/2$; $t = -\sqrt{3}$.

$rt - s^2 = 3 - 3/4 = 9/4 > 0$ and $r < 0 \Rightarrow$ MAXIMUM.

$$f_{max} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

Problem 36: divide 24 into three parts so that the product of the parts is a maximum.

Let the parts be $x, y, 24 - x - y$; $f = xy(24 - x - y)$. $f_x = y(24 - 2x - y) = 0$, $f_y = x(24 - x - 2y) = 0 \Rightarrow 2x + y = 24$, $x + 2y = 24 \Rightarrow x = y = 8$. $r = -2y = -16$, $s = 24 - 2x - 2y = -8$, $t = -2x = -16$: $rt - s^2 = 256 - 64 > 0$, $r < 0 \Rightarrow$ maximum.

parts 8, 8, 8; maximum product = 512

Problem 37: find the maximum of $f = xy(a - x - y)$.

$f_x = y(a - 2x - y) = 0$, $f_y = x(a - x - 2y) = 0 \Rightarrow x = y = a/3$ (also $(0,0)$, $(a,0)$, $(0,a)$ – saddles). $r = -2a/3$, $s = -a/3$, $t = -2a/3 \Rightarrow rt - s^2 = a^2/3 > 0$, $r < 0 \Rightarrow$ MAX. $f_{max} = (a/3)(a/3)(a/3) = a^3/27$.

8. Lagrange's Method of Multipliers

To find the extreme values of $f(x, y, z)$ subject to a constraint $\varphi(x, y, z) = 0$:

1. Form $F = f + \lambda\varphi$ (λ = Lagrange multiplier).
2. Solve $F_x = 0$, $F_y = 0$, $F_z = 0$ together with $\varphi = 0$ for x, y, z, λ.
3. Substitute the point(s) in f to get the extreme value; nature from the geometry of the problem.
4. Two constraints $\varphi_1 = 0$, $\varphi_2 = 0$: $F = f + \lambda_1\varphi_1 + \lambda_2\varphi_2$.

Problem 38: find the minimum value of $x^2 + y^2 + z^2$ subject to $x + y + z = 3a$.

$F = x^2 + y^2 + z^2 + \lambda(x + y + z - 3a)$. $F_x = 2x + \lambda = 0$, $F_y = 2y + \lambda = 0$, $F_z = 2z + \lambda = 0 \Rightarrow x = y = z = -\lambda/2$. With the constraint: $3x = 3a \Rightarrow x = y = z = a$.

$$\text{minimum value} = a^2 + a^2 + a^2 = 3a^2$$

Problem 39: find the shortest distance from the origin to the plane $3x + 2y + z = 12$.

Minimise $f = x^2 + y^2 + z^2$ with $\varphi = 3x + 2y + z - 12 = 0$. $2x + 3\lambda = 0$, $2y + 2\lambda = 0$, $2z + \lambda = 0 \Rightarrow x = -3\lambda/2$, $y = -\lambda$, $z = -\lambda/2$. Substituting: $3(-3\lambda/2) + 2(-\lambda) - \lambda/2 = 12 \Rightarrow -7\lambda = 12 \Rightarrow \lambda = -12/7$.

$x = 18/7$, $y = 12/7$, $z = 6/7$; $f = (324 + 144 + 36)/49 = 504/49 = 72/7$.

$$\text{shortest distance} = \sqrt{(72/7)} = 12/\sqrt{14} \approx 3.207$$

Problem 40: find the maximum volume of a rectangular box inscribed in the sphere $x^2 + y^2 + z^2 = a^2$.

Vertex (x, y, z) in the first octant; $V = 8xyz$. $F = xyz + \lambda(x^2 + y^2 + z^2 - a^2)$: $yz + 2\lambda x = 0$, $zx + 2\lambda y = 0$, $xy + 2\lambda z = 0$. Multiply by x, y, z: $xyz = -2\lambda x^2 = -2\lambda y^2 = -2\lambda z^2 \Rightarrow x = y = z = a/\sqrt{3}$.

$$V_{max} = 8\,(a/\sqrt{3})^3 = \frac{8a^3}{3\sqrt{3}}$$

Problem 41: find the dimensions of an open rectangular box of volume 32 cm^3 with the least surface area.

$S = xy + 2z(x + y)$ with $xyz = 32$. $F = xy + 2yz + 2zx + \lambda(xyz - 32)$. $y + 2z + \lambda yz = 0$, $x + 2z + \lambda xz = 0$, $2y + 2x + \lambda xy = 0$. Multiply the 1st by x, 2nd by y, 3rd by z: $xy + 2xz + \lambda xyz = yx + 2yz + \lambda xyz \Rightarrow x = y$; and with the 3rd: $2xz + \ldots \Rightarrow x = y = 2z$.

$(2z)(2z)z = 32 \Rightarrow z^3 = 8 \Rightarrow z = 2$, $x = y = 4$. Least surface area $S = 16 + 2 \cdot 2 \cdot 8 = 48\ cm^2$.

Problem 42: find the largest volume of a rectangular parallelepiped inscribed in the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$.

$V = 8xyz$; $F = xyz + \lambda(x^2/a^2 + y^2/b^2 + z^2/c^2) \Rightarrow yz + 2\lambda x/a^2 = 0$ etc.

Multiplying by x, y, z and equating: $x^2/a^2 = y^2/b^2 = z^2/c^2 = 1/3 \Rightarrow x = a/\sqrt{3}$, $y = b/\sqrt{3}$, $z = c/\sqrt{3}$.

$$V_{max} = 8 \cdot \frac{abc}{3\sqrt{3}} = \frac{8abc}{3\sqrt{3}}$$

Problem 43: find the points on $z^2 = xy + 1$ nearest to the origin.

Minimise $x^2 + y^2 + z^2$ subject to $z^2 - xy - 1 = 0$. F: $2x - \lambda y = 0$, $2y - \lambda x = 0$, $2z + 2\lambda z = 0$. If $z \neq 0$ then $\lambda = -1 \Rightarrow 2x + y = 0$ and $2y + x = 0 \Rightarrow x = y = 0 \Rightarrow z^2 = 1$. Points $(0, 0, \pm 1)$, minimum distance 1.

Problem 44: maximise $x^2y^3z^4$ subject to $2x + 3y + 4z = a$.

Maximise $\log f = 2\log x + 3\log y + 4\log z$: $2/x + 2\lambda = 0$, $3/y + 3\lambda = 0$, $4/z + 4\lambda = 0 \Rightarrow x = y = z = -1/\lambda$. Constraint: $9x = a \Rightarrow x = y = z = a/9$.

$$f_{max} = (a/9)^2 (a/9)^3 (a/9)^4 = \frac{a^9}{9^9}$$

Important Questions – Unit IV

1. Test continuity of $f(x, y)$ at the origin; explain differentiability.
2. Find partial derivatives; verify $f_{xy} = f_{yx}$; Euler's theorem problems.
3. Total derivative and chain rule problems; implicit differentiation.
4. Directional derivative and direction of maximum rate of change.
5. Taylor / Maclaurin series of a function of two variables.
6. Jacobians; show functional dependence and find the relation.
7. Maxima and minima of functions of two variables ($rt - s^2$ test).
8. Lagrange multipliers: extreme values with one / two constraints; box and distance problems.

LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - V

Multiple Integrals

Topics covered:

- Double integrals and triple integrals
- Change of order of integration
- Change of variables: polar, cylindrical and spherical coordinates
- Areas by double integrals
- Volumes by double and triple integrals

Handwritten Notes - B.Tech (JNTUA R23)

1. Double Integrals

The double integral of $f(x, y)$ over a region R is the limit of $\Sigma f(x_i, y_i) \Delta A$: $\iint_R f(x, y)\, dA$. It is evaluated as an ITERATED (repeated) integral.

$$\int_{x=a}^{b} \int_{y=\varphi_1(x)}^{\varphi_2(x)} f(x, y)\, dy\, dx \quad (y\text{-limits depend on } x : \text{vertical strip})$$

$$\int_{y=c}^{d} \int_{x=\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy \quad (x\text{-limits depend on } y : \text{horizontal strip})$$

- Integrate the INNER integral first, treating the outer variable as a constant.
- Constant limits (rectangle) – the order does not matter (Fubini).
- For a region R draw a rough sketch first, the limits come from the sketch.

Problem 1: evaluate $\int_0^1 \int_0^2 (x^2 + y^2)\, dy\, dx$.

$$\text{inner} : \int_0^2 (x^2 + y^2)\, dy = \left[x^2 y + y^3/3 \right]_0^2 = 2x^2 + 8/3$$

$$\boxed{\text{outer} : \int_0^1 (2x^2 + 8/3)\, dx = 2/3 + 8/3 = 10/3}$$

Problem 2: evaluate $\iint xy\, dx\, dy$ over the region bounded by the x-axis, the ordinate $x = 2a$ and the curve $x^2 = 4ay$.

For a given x, y runs from 0 to $x^2/(4a)$; x runs from 0 to $2a$.

$$\boxed{\int_0^{2a} x \left[y^2/2 \right]_0^{x^2/4a} dx = \int_0^{2a} \frac{x^5}{32a^2}\, dx = \frac{(2a)^6}{6 \cdot 32a^2} = \frac{a^4}{3}}$$

Problem 3: evaluate $\iint xy(x + y)\, dx\, dy$ over the area between $y = x^2$ and $y = x$.

The curves meet where $x^2 = x$: $x = 0, 1$. For $0 \le x \le 1$, y goes from x^2 (lower) to x (upper).

$$\int_0^1 \int_{x^2}^{x} (x^2 y + x y^2)\, dy\, dx = \int_0^1 \left[x^2 y^2/2 + x y^3/3 \right]_{x^2}^{x} dx$$

$$= \int_0^1 \left(\frac{5x^4}{6} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$$

$$= \frac{1}{6} - \frac{1}{14} - \frac{1}{24} = \frac{3}{56}$$

Problem 4: evaluate $\iint e^{x+y}\, dx\, dy$ over the triangle bounded by $x = 0$, $y = 0$, $x + y = 1$.

$$\int_0^1 \int_0^{1-x} e^{x+y}\, dy\, dx = \int_0^1 (e - e^x)\, dx = e - (e - 1) = 1$$

Problem 5: evaluate $\iint xy\, dx\, dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$.

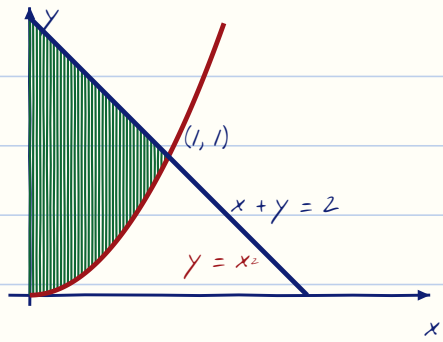
$$\int_0^a x \cdot \frac{a^2 - x^2}{2}\, dx = \frac{1}{2}\left[\frac{a^2x^2}{2} - \frac{x^4}{4} \right]_0^a = \frac{a^4}{8}$$

2. Change of Order of Integration

If the limits are such that one order is difficult (or impossible) we sketch the region and re-describe it with the other variable first.

1. Draw the region from the given limits (outer limits are numbers, inner limits are curves).
2. Draw a strip in the OTHER direction, note where it enters and leaves the region ⇒ new inner limits; the extreme values give the new outer limits.
3. If the strip enters/leaves through different curves in different parts, SPLIT the region.
4. Evaluate in the new order – the value is unchanged.

Problem 6: change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ and evaluate.



Region of integration:

$0 \le x \le 1$, $x^2 \le y \le 2 - x$

dy first (vertical strip)

The region lies between the parabola $y = x^2$ and the line $y = 2 - x$ for $0 \le x \le 1$ (they meet at $(1, 1)$). Draw HORIZONTAL strips:

- For $0 \le y \le 1$: x runs from 0 to $\sqrt{y}$.
- For $1 \le y \le 2$: x runs from 0 to $2 - y$.

$$I = \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy = \int_0^1 \frac{y^2}{2} \, dy + \int_1^2 \frac{y(2-y)^2}{2} \, dy$$

$$= \frac{1}{6} + \frac{5}{24} = \frac{3}{8}$$

(Check: the original order gives $\int_0^1 x[(2-x)^2 - x^4]/2 \, dx = 3/8$ ✓.)

Problem 7: change the order and evaluate $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} dy \, dx$ (the area between two parabolas).

The curves $y = x^2/4$ and $y = 2\sqrt{x}$ (i.e. $x = y^2/4 \ldots y^2 = 4x$) meet at $(0, 0)$ and $(4, 4)$. Horizontal strip: x from $y^2/4$ to $2\sqrt{y}$, $0 \le y \le 4$.

$$\int_0^4 \int_{y^2/4}^{2\sqrt{y}} dx \, dy = \int_0^4 \left(2\sqrt{y} - \frac{y^2}{4} \right) dy = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$$

Problem 8: evaluate by changing the order: $\int_0^\infty \int_x^\infty (e^{-y}/y) \, dy \, dx$.

Region: $0 \le x < \infty$, $x \le y < \infty$, i.e. $0 \le y < \infty$ and $0 \le x \le y$.

Interchanging:

$$\int_0^\infty \frac{e^{-y}}{y} \left(\int_0^y dx \right) dy = \int_0^\infty \frac{e^{-y}}{y} \cdot y \, dy = \int_0^\infty e^{-y} \, dy = 1$$

Problem 9: evaluate $\int_0^1 \int_x^1 (\sin y / y)\, dy\, dx$ (cannot be done in the given order).

($\sin y/y$ has no elementary antiderivative.) Region: $0 \le x \le 1,\ x \le y \le 1 \Rightarrow 0 \le y \le 1,\ 0 \le x \le y$.

$$\int_0^1 \frac{\sin y}{y} \left(\int_0^y dx \right) dy = \int_0^1 \sin y\, dy = 1 - \cos 1$$

Problem 10: change the order of $\int_0^a \int_{x^2/a}^{2a-x} xy\, dy\, dx$ and evaluate.

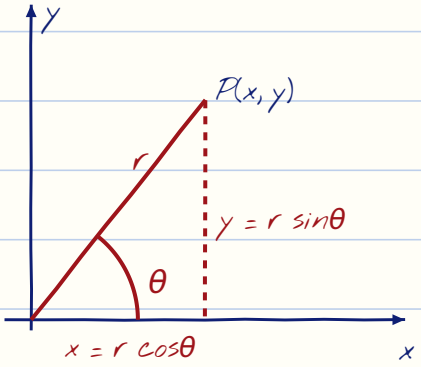
Parabola $x^2 = ay$ and line $x + y = 2a$ meet at (a, a). For $0 \le y \le a$: x from 0 to $\sqrt{(ay)}$; for $a \le y \le 2a$: x from 0 to $2a - y$.

$$I = \int_0^a \int_0^{\sqrt{(ay)}} xy\, dx\, dy + \int_a^{2a} \int_0^{2a-y} xy\, dx\, dy = \frac{a^4}{6} + \frac{5a^4}{24} = \frac{3a^4}{8}$$

3. Change of Variables – Polar Coordinates

For a double integral, a substitution $x = x(u, v),\ y = y(u, v)$ gives

$$\iint f(x, y)\, dx\, dy = \iint f(x(u, v), y(u, v))\, |J|\, du\, dv, \quad J = \frac{\partial(x, y)}{\partial(u, v)}$$



$x = r\cos\theta,\ y = r\sin\theta$

$x^2 + y^2 = r^2$

$dx\, dy = r\, dr\, d\theta$

(Jacobian = r)

$$\iint f(x, y)\, dx\, dy = \iint f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$$

Use polar form when the region is a circle / sector / part of a circle or the integrand contains $x^2 + y^2$. For a circle through the origin $r = 2a\cos\theta$ use $-\pi/2 \le \theta \le \pi/2$; circle at the origin: $0 \le r \le a,\ 0 \le \theta \le 2\pi$.

Problem 11: evaluate $\iint e^{-(x^2+y^2)}\,dx\,dy$ over the first quadrant, hence find $\int_0^\infty e^{-x^2}\,dx$.

Put $x = r\cos\theta, y = r\sin\theta$:

$$I = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\theta = \int_0^{\pi/2}\left[-\frac{e^{-r^2}}{2}\right]_0^\infty d\theta$$

$$= \int_0^{\pi/2}\frac{1}{2}\,d\theta = \frac{\pi}{4}$$

But $I = \left(\int_0^\infty e^{-x^2}\,dx\right)^2$ (the integral separates in x and y), so

$$\boxed{\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}}$$

Problem 12: evaluate $\int_0^a \int_0^{\sqrt{(a^2-x^2)}} (x^2+y^2)\,dy\,dx$.

Region: quarter circle $x^2+y^2 \le a^2$ in the first quadrant $\Rightarrow 0 \le r \le a, 0 \le \theta \le \pi/2$.

$$\boxed{\int_0^{\pi/2}\int_0^a r^2 \cdot r\,dr\,d\theta = \frac{a^4}{4}\cdot\frac{\pi}{2} = \frac{\pi a^4}{8}}$$

Problem 13: evaluate $\int_0^2 \int_0^{\sqrt{(2x-x^2)}} x\,dy\,dx$.

$y^2 \le 2x - x^2 \Rightarrow (x-1)^2 + y^2 \le 1$: upper half of the circle with centre (1, 0), $r = 2\cos\theta$, $0 \le \theta \le \pi/2$.

$$\int_0^{\pi/2}\int_0^{2\cos\theta} (r\cos\theta)\, r\,dr\,d\theta = \int_0^{\pi/2}\cos\theta\cdot\frac{8\cos^3\theta}{3}\,d\theta$$

$$\boxed{= \frac{8}{3}\int_0^{\pi/2}\cos^4\theta\,d\theta = \frac{8}{3}\cdot\frac{3\pi}{16} = \frac{\pi}{2}}$$

Problem 14: evaluate $\iint dx\,dy/(1+x^2+y^2)$ over the circle $x^2+y^2 \le a^2$.

$$\int_0^{2\pi}\int_0^{a}\frac{r}{1+r^2}\,dr\,d\theta = 2\pi\cdot\frac{1}{2}\log(1+a^2) = \pi\log(1+a^2)$$

Problem 15 (general substitution): evaluate $\iint (x^2+y^2)\,dx\,dy$ over the region $|x|+|y|\le 1$ using $u=x+y$, $v=x-y$.

$x=(u+v)/2$, $y=(u-v)/2$; $J=\partial(x,y)/\partial(u,v)=-1/2 \Rightarrow |J|=1/2$. Region: $-1\le u\le 1$, $-1\le v\le 1$. $x^2+y^2=(u^2+v^2)/2$.

$$I=\int_{-1}^{1}\int_{-1}^{1}\frac{u^2+v^2}{2}\cdot\frac{1}{2}\,du\,dv=\frac{1}{4}\left[2\cdot\frac{2}{3}\cdot 2\right]=\frac{2}{3}$$

4. Triple Integrals

$\iiint_V f(x,y,z)\,dV=\int_{x1}^{x2}\int_{y_1(x)}^{y_2(x)}\int_{z_1(x,y)}^{z_2(x,y)} f\,dz\,dy\,dx$. Integrate the INNERMOST variable first. If $f=1$ the integral gives the VOLUME of V.

Problem 16: evaluate $\int_0^a\int_0^b\int_0^c (x^2+y^2+z^2)\,dz\,dy\,dx$.

$$=\int_0^a\int_0^b\left[(x^2+y^2)c+\frac{c^3}{3}\right]dy\,dx$$

$$=\int_0^a\left[x^2bc+\frac{b^3c}{3}+\frac{bc^3}{3}\right]dx=\frac{abc(a^2+b^2+c^2)}{3}$$

Problem 17: evaluate $\iiint x\,y\,z\,dx\,dy\,dz$ over the tetrahedron $x\ge 0$, $y\ge 0$, $z\ge 0$, $x+y+z\le 1$.

$$\int_0^1\int_0^{1-x}\int_0^{1-x-y} xyz\,dz\,dy\,dx=\int_0^1\int_0^{1-x} xy\,\frac{(1-x-y)^2}{2}\,dy\,dx$$

$$=\int_0^1 x\,\frac{(1-x)^4}{24}\,dx=\frac{1}{720}$$

Problem 18: find $\iiint x\,dV$ and the volume of the same tetrahedron $x + y + z \le 1$ (first octant).

Volume $= \int_0^1 \int_0^{1-x} (1 - x - y)\,dy\,dx = \int_0^1 (1 - x)^2/2\,dx = 1/6$. $\iiint x\,dV = \int_0^1 x(1 - x)^2/2\,dx = 1/24$.

Problem 19: evaluate $\iiint (x + y + z)\,dV$ over the cube $0 \le x, y, z \le 1$.

By symmetry each of x, y, z contributes $\iiint x\,dV = 1/2$: total $= 3/2$.

Cylindrical coordinates: $x = r\cos\theta, y = r\sin\theta, z = z$; $dV = r\,dr\,d\theta\,dz$. Use for cylinders, cones, paraboloids with z-axis symmetry.

Spherical coordinates: $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi, z = r\cos\theta$; $dV = r^2\sin\theta\,dr\,d\theta\,d\varphi$ ($0 \le r, 0 \le \theta \le \pi, 0 \le \varphi \le 2\pi$). Use for spheres, cones about the z-axis.

System	Variables	dV	$x^2 + y^2 + z^2$
Cartesian	x, y, z	$dx\,dy\,dz$	$x^2 + y^2 + z^2$
Cylindrical	r, θ, z	$r\,dr\,d\theta\,dz$	$r^2 + z^2$
Spherical	r, θ, φ	$r^2\sin\theta\,dr\,d\theta\,d\varphi$	r^2

Problem 20: evaluate $\iiint z\,dV$ over the cylinder $x^2 + y^2 \le a^2, 0 \le z \le h$ (cylindrical coordinates).

$$\int_0^{2\pi}\int_0^a\int_0^h z \cdot r\,dz\,dr\,d\theta = 2\pi \cdot \frac{a^2}{2} \cdot \frac{h^2}{2} = \frac{\pi a^2 h^2}{2}$$

Problem 21: evaluate $\iiint (x^2 + y^2 + z^2)\,dV$ over the sphere $x^2 + y^2 + z^2 \le 1$.

$$\int_0^{2\pi}\int_0^{\pi}\int_0^1 r^2 \cdot r^2\sin\theta\,dr\,d\theta\,d\varphi = \frac{1}{5} \cdot 2 \cdot 2\pi = \frac{4\pi}{5}$$

Problem 22: evaluate $\iiint xyz\,dV$ over the positive octant of the sphere $x^2 + y^2 + z^2 \le a^2$.

$$\int_0^a r^5\,dr \cdot \int_0^{\pi/2} \sin^3\theta\cos\theta\,d\theta \cdot \int_0^{\pi/2} \sin\varphi\cos\varphi\,d\varphi$$

$$= \frac{a^6}{6} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{a^6}{48}$$

5. Areas by Double Integrals

$$\text{Area of } R = \iint_R dx\,dy \quad (\text{polar}: \iint r\,dr\,d\theta)$$

Problem 23: find the area bounded by the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

They meet at $(0, 0)$ and $(4a, 4a)$. For $0 \le x \le 4a$: y from $x^2/4a$ to $2\sqrt{(ax)}$.

$$A = \int_0^{4a} \left(2\sqrt{a}\,\sqrt{x} - \frac{x^2}{4a} \right) dx$$

$$= \left[\frac{4\sqrt{a}}{3} x^{3/2} - \frac{x^3}{12a} \right]_0^{4a} = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3}$$

Problem 24: find the area of the ellipse $x^2/a^2 + y^2/b^2 = 1$.

$A = 4\int_0^a \int_0^{(b/a)\sqrt{(a^2-x^2)}} dy\,dx = (4b/a)\int_0^a \sqrt{(a^2 - x^2)}\,dx = (4b/a)(\pi a^2/4)$:

$$A = \pi a b$$

Problem 25: find the area of the cardioid $r = a(1 + \cos\theta)$.

$$A = 2\int_0^{\pi} \int_0^{a(1+\cos\theta)} r\,dr\,d\theta = a^2 \int_0^{\pi} (1 + \cos\theta)^2\,d\theta$$

$$= a^2 \left(\pi + 0 + \pi/2 \right) = \frac{3\pi a^2}{2}$$

Problem 26: find the area enclosed by $y = x$ and $y = x^2$. And the area of the circle $r = a$.

$\int_0^1 (x - x^2)\,dx = 1/2 - 1/3 = 1/6$. Circle: $\int_0^{2\pi} \int_0^a r\,dr\,d\theta = 2\pi \cdot a^2/2 = \pi a^2$.

Problem 27: find the total area of the lemniscate $r^2 = a^2 \cos 2\theta$.

$$A = 4\int_0^{\pi/4}\int_0^{a\sqrt{\cos 2\theta}} r\,dr\,d\theta = 2a^2\int_0^{\pi/4}\cos 2\theta\,d\theta = 2a^2\cdot\frac{1}{2} = a^2$$

Problem 28: find the area inside the circle $r = 2a\cos\theta$ and outside $r = a$.

They meet where $2a\cos\theta = a \Rightarrow \theta = \pm\pi/3$.

$$A = 2\int_0^{\pi/3}\int_a^{2a\cos\theta} r\,dr\,d\theta = \int_0^{\pi/3}(4a^2\cos^2\theta - a^2)\,d\theta$$

$$= a^2[\theta + \sin 2\theta]_0^{\pi/3}$$

$$= a^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right)$$

6. Volumes by Double and Triple Integrals

$$\text{Volume} = \iint_R (z_{upper} - z_{lower})\,dx\,dy$$

$$\text{Volume} = \iiint_V dx\,dy\,dz$$

Problem 29: find the volume of the sphere of radius a (double integral).

$V = 2\iint\sqrt{(a^2 - x^2 - y^2)}\,dx\,dy$ over the circle $x^2 + y^2 \le a^2$ (upper half-sphere doubled). Polar:

$$V = 2\int_0^{2\pi}\int_0^a \sqrt{(a^2 - r^2)}\,r\,dr\,d\theta = 2\cdot 2\pi\cdot\frac{a^3}{3} = \frac{4\pi a^3}{3}$$

Problem 30: find the volume of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

$$V = \iint (4 - r^2)\, r\, dr\, d\theta = 2\pi \left[2r^2 - \frac{r^4}{4} \right]_0^2 = 2\pi (8 - 4) = 8\pi$$

Problem 31: find the volume of the tetrahedron bounded by the coordinate planes and $x + y + z = 1$ using a double integral.

$V = \iint (1 - x - y)\, dy\, dx$ over $0 \le x \le 1, 0 \le y \le 1 - x = \int_0^1 (1 - x)^2/2\, dx = 1/6$. (For $x/a + y/b + z/c = 1$: $V = abc/6$.)

Problem 32: find the volume of the solid common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.

In the first octant $z = \sqrt{(a^2 - x^2)}$ over the quarter-circle $0 \le y \le \sqrt{(a^2 - x^2)}$, $0 \le x \le a$; total = 8 × first octant.

$$V = 8 \int_0^a \int_0^{\sqrt{(a^2 - x^2)}} \sqrt{(a^2 - x^2)}\, dy\, dx$$

$$= 8 \int_0^a (a^2 - x^2)\, dx = 8 \cdot \frac{2a^3}{3} = \frac{16a^3}{3}$$

Problem 33: find the volume of the cone $z = \sqrt{(x^2 + y^2)}$ cut off by the plane $z = h$ (cylindrical coordinates).

$$V = \int_0^{2\pi} \int_0^h \int_r^h r\, dz\, dr\, d\theta = 2\pi \int_0^h r(h - r)\, dr = 2\pi \frac{h^3}{6} = \frac{\pi h^3}{3}$$

Problem 34: find the volume of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ (triple integral).

Put $x = a\,u, y = b\,v, z = c\,w$ (Jacobian = abc): the ellipsoid becomes the unit sphere $u^2 + v^2 + w^2 \le 1$ whose volume is $4\pi/3$.

$$V = abc \iiint_{\text{unit sphere}} du\, dv\, dw = \frac{4\pi abc}{3}$$

Problem 35: find the volume of the solid bounded by $z = 0$ and $z = 1 - x^2 - y^2$.

$$\boxed{V = \int_0^{2\pi} \int_0^1 (1 - r^2)\, r\, dr\, d\theta = 2\pi \left[\frac{1}{2} - \frac{1}{4} \right] = \frac{\pi}{2}}$$

Problem 36: find the volume of the part of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ax$.

The cylinder is the circle $r = a\cos\theta$ in polar form; the sphere gives $z = \pm\sqrt{(a^2 - r^2)}$. By symmetry $V = 4 \times$ (first quadrant part):

$$V = 4 \int_0^{\pi/2} \int_0^{a\cos\theta} \sqrt{(a^2 - r^2)}\, r\, dr\, d\theta$$

$$= \frac{4a^3}{3} \int_0^{\pi/2} (1 - \sin^3\theta)\, d\theta = \frac{4a^3}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right)$$

$$\boxed{V = \frac{2a^3}{9} (3\pi - 4)}$$

Summary of useful results:

Quantity	Value
Volume of sphere / hemisphere	$4\pi a^3/3$ / $2\pi a^3/3$
Volume of cylinder (radius a, height h)	$\pi a^2 h$
Volume of cone (radius a, height h)	$\pi a^2 h/3$
Area of circle / ellipse	πa^2 / πab
Jacobians	polar r; cylindrical r; spherical $r^2 \sin\theta$
$\int_0^\infty e^{-x^2}\, dx$	$\sqrt{\pi}/2$

Important Questions - Unit V

1. Evaluate double integrals over given regions (rectangle, triangle, curves).
2. Change the order of integration and evaluate (problems 6-10).
3. Evaluate integrals by changing to polar coordinates.
4. Evaluate triple integrals, use cylindrical and spherical coordinates.
5. Find areas by double integration (parabolas, ellipse, cardioid, lemniscate).
6. Find volumes by double / triple integrals (sphere, paraboloid, cone, cylinders, ellipsoid, tetrahedron).